FATIMA JINNAH WOMEN UNIVERSITY

RAWALPINDI

Department Of Software Engineering

Cloud Computing Lab

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Section: 5-B

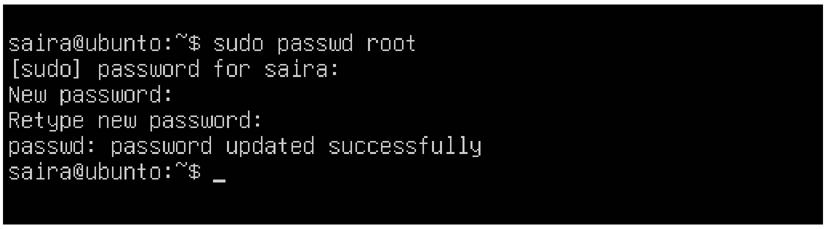
LAB NO 6

Lab Task:

Task 1 – Switch to root with su - and back to a normal user

1. Set a root password (Ubuntu root is disabled by default; this enables su - temporarily for the lab):

sudo passwd root

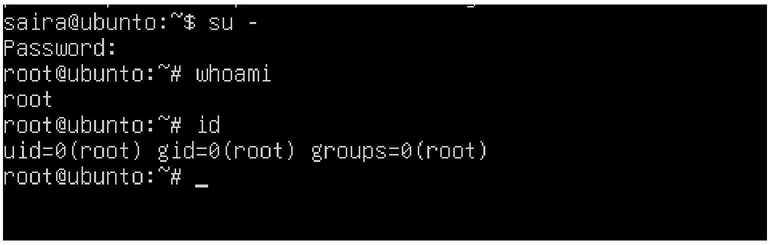


```
saira@ubunto:~$ sudo passwd root
[sudo] password for saira:
New password:
Retype new password:
passwd: password updated successfully
saira@ubunto:~$ _
```

Enter a temporary root password for the lab

2. Switch to root and verify:

su -
whoami
id



```
saira@ubunto:~$ su -
Password:
root@ubunto:~# whoami
root
root@ubunto:~# id
uid=0(root) gid=0(root) groups=0(root)
root@ubunto:~# _
```

3. Switch back to your normal user:

exit

whoami

```
root@ubunto:~# exit
logout
saira@ubunto:~$ whoami
saira
saira@ubunto:~$
```

Task 2 – Create user tom and verify in passwd/group/shadow

Goal: Create a user named tom, then verify the account in system files.

1. Create user tom (interactive, sets password and home directory):

sudo adduser tom

```
saira@ubunto:~$ sudo adduser tom
info: Adding user `tom' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `tom' (1001) ...
info: Adding new user `tom' (1001) with group `tom (1001)' ...
info: Creating home directory `/home/tom' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for tom
Enter the new value, or press ENTER for the default
        Full Name []: Tom cruise
        Room Number []: 200
        Work Phone []: 32455-432
        Home Phone []: 12345-543
        Other []: n/a
Is the information correct? [Y/n] y
info: Adding new user `tom' to supplemental / extra groups `users' ...
info: Adding user `tom' to group `users' ...
saira@ubunto:~$
```

2. Verify tom in system files (view and visually confirm presence):

cat /etc/passwd

```
info: Adding user `tom' to group `users' ...
saira@ubunto:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uuidd:x:104:105::/run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
saira:x:1000:1000:Saira:/home/saira:/bin/bash
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
rtkit:x:110:110:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
cups-pk-helper:x:111:111:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
lightdm:x:112:115:Light Display Manager:/var/lib/lightdm:/bin/false
whoopsie:x:113:117::/nonexistent:/bin/false
avahi:x:114:118:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
pulse:x:115:119:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
saned:x:116:122::/var/lib/saned:/usr/sbin/nologin
cups-browsed:x:117:111::/nonexistent:/usr/sbin/nologin
colord:x:118:123:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
xrdp:x:119:124::/run/xrdp:/usr/sbin/nologin
tom:x:1001:1001:Tom cruise,200,32455-432,12345-543,n/a:/home/tom:/bin/bash
saira@ubunto:~$
```

cat /etc/group

```
list:x:38:
irc:x:39:
src:x:40:
shadow:x:42:
utmp:x:43:
video:x:44:
sasl:x:45:
plugdev:x:46:saira
staff:x:50:
games:x:60:
users:x:100:tom
nogroup:x:65534:
systemd-journal:x:999:
systemd-network:x:998:
systemd-timesync:x:997:
input:x:996:
sgx:x:995:
kvm:x:994:
render:x:993:
lxd:x:101:saira
messagebus:x:102:
systemd-resolve:x:992:
_ssh:x:103:
polkitd:x:991:
crontab:x:990:
syslog:x:104:
uuidd:x:105:
rdma:x:106:
tcpdump:x:107:
tss:x:108:
landscape:x:109:
fwupd-refresh:x:989:
saira:x:1000:
rtkit:x:110:
lpadmin:x:111:
netdev:x:112:
bluetooth:x:113:
ssl-cert:x:114:
lightdm:x:115:
nopasswdlogin:x:116:
whoopsie:x:117:
avahi:x:118:
pulse:x:119:
pulse-access:x:120:
scanner:x:121:saned
saned:x:122:
colord:x:123:
xrdp:x:124:
tom:x:1001:
saira@ubunto:~$
```

sudo cat /etc/shadow

```
saira@ubunto:~$ sudo cat /etc/shadow
root:$y$j9T$4bT2ZpeLSow90eNP31WBQ.$HBTexwxjIID/9gH0aBe5FcEbC.w.DzWeQ.9BKQBzA0D:20398:0:99999:7:::
daemon:*:20305:0:99999:7:::
bin:*:20305:0:99999:7:::
sys:*:20305:0:99999:7:::
sync:*:20305:0:99999:7:::
games:*:20305:0:99999:7:::
man:*:20305:0:99999:7:::
lp:*:20305:0:99999:7:::
mail:*:20305:0:99999:7:::
news:*:20305:0:99999:7:::
uucp:*:20305:0:99999:7:::
proxy:*:20305:0:99999:7:::
www-data:*:20305:0:99999:7:::
backup:*:20305:0:99999:7:::
list:*:20305:0:99999:7:::
irc:*:20305:0:99999:7:::
_apt:*:20305:0:99999:7:::
nobody:*:20305:0:99999:7:::
systemd-network:!*:20305::::::
systemd-timesync:!*:20305::::::
dhcpcd:!:20305::::::
messagebus:!:20305::::::
systemd-resolve:!*:20305::::::
pollinate:!:20305::::::
polkitd:!*:20305::::::
syslog:!:20305::::::
uuidd:!:20305::::::
tcpdump:!:20305::::::
tss:!:20305::::::
landscape:!:20305::::::
fwupd-refresh:!*:20305::::::
usbmux:!:20395::::::
saira:$6$eCpcKBfqODD.C6MZ$U5hV54rnmnNTEoZNH3uoJ..3.enaNcVOGcH30ChkbsUuNpZMhedPRUuZkWBQ/MkAt3oyVFSdGo6eSvhd.DR9Y.:20395:0:99999:7:::
sshd:!:20396::::::
rtkit:!:20396::::::
dnsmasq:!:20396::::::
cups-pk-helper:!:20396::::::
lightdm:!:20396::::::
whoopsie:!:20396::::::
avahi:!:20396::::::
pulse:!:20396::::::
saned:!:20396::::::
cups-browsed:!:20396::::::
colord:!:20396::::::
xrdp:!:20396::::::
tom:$y$j9T$OjpXyswseiyfs/143WOcT.$CIf8Ck1OdvXOo5/lcaJW7jl5z2h6F9L06JwzlDG89T.:20398:0:99999:7:::
saira@ubunto:~$
```

Task 3 – Create groups; change tom's primary and secondary groups

Goal: Create groups developer, devops, and designer. Change tom's primary group and manage secondary groups.

1. Create groups and verify by viewing /etc/group (visually confirm entries exist):

sudo groupadd developer
sudo groupadd devops
sudo groupadd designer

```
saira@ubunto:~$ sudo groupadd developer
saira@ubunto:~$ sudo groupadd devops
saira@ubunto:~$ sudo groupadd designer
saira@ubunto:~$
```

cat /etc/group

```
systemd-journal:x:999:
systemd-network:x:998:
systemd-timesync:x:997:
input:x:996:
sgx:x:995:
kvm:x:994:
render:x:993:
lxd:x:101:saira
messagebus:x:102:
systemd-resolve:x:992:
_ssh:x:103:
polkitd:x:991:
crontab:x:990:
syslog:x:104:
uuidd:x:105:
rdma:x:106:
tcpdump:x:107:
tss:x:108:
landscape:x:109:
fwupd-refresh:x:989:
saira:x:1000:
rtkit:x:110:
lpadmin:x:111:
netdev:x:112:
bluetooth:x:113:
ssl-cert:x:114:
lightdm:x:115:
nopasswdlogin:x:116:
whoopsie:x:117:
avahi:x:118:
pulse:x:119:
pulse-access:x:120:
scanner:x:121:saned
saned:x:122:
colord:x:123:
xrdp:x:124:
tom:x:1001:
developer:x:1002:
devops:x:1003:
designer:x:1004:
saira@ubunto:~$
```

2. Change tom's primary group to designer and verify:

sudo usermod -g designer tom

id tom

```
saira@ubunto:~$ sudo usermod -g designer tom
saira@ubunto:~$ id tom
uid=1001(tom) gid=1004(designer) groups=1004(designer),100(users)
saira@ubunto:~$ _
```

3. Add secondary groups developer and devops to tom and verify:

sudo usermod -aG developer,devops tom

id tom

groups tom

```
saira@ubunto:~$ sudo usermod -aG developer,devops tom
saira@ubunto:~$ id tom
uid=1001(tom) gid=1004(designer) groups=1004(designer),100(users),1002(developer),1003(devops)
saira@ubunto:~$ groups tom
tom : designer users developer devops
saira@ubunto:~$
```

4. Replace all secondary groups so only tom (user's own group) remains and verify:

sudo usermod -G tom tom

id tom

groups tom

```
saira@ubuntu:~$ sudo usermod -G tom tom
saira@ubuntu:~$ id tom
uid=1001(tom) gid=1004(designer) groups=1004(designer),1001(tom)
saira@ubuntu:~$ groups tom
tom : designer tom
saira@ubuntu:~$
```

Task 4 – Create/delete users (Jerry, Scooby) and groups (jolly, anime)

Goal: Create users using both adduser and useradd, demonstrate login/password/home directory differences, then delete users/groups.

1. Create users:

sudo adduser Jerry

sudo useradd Scooby

```
saira@ubuntu:~$ sudo adduser Jerry
err: Please enter a username matching the regular expression
          configured via the NAME_REGEX configuration variable.  Use the
          `--allow-bad-names' option to relax this check or reconfigure
          NAME_REGEX in configuration.
saira@ubuntu:~$ sudo useradd Scooby
```

2. Try to log in as Scooby immediately (expected authentication failure because there is no password yet):

su – Scooby

```
          NAME_REGEX in configuration.
saira@ubuntu:~$ su - Scooby
Password:
su: Authentication failure
```

3. Set a password for Scooby:

sudo passwd Scooby

```
saira@ubuntu:~$ sudo passwd Scooby
New password:
Retype new password:
passwd: password updated successfully
saira@ubuntu:~$ _
```

4. Try logging in as Scooby again (home directory still missing; expect a message such as “No directory, logging in with HOME=/”):

su – Scooby

```
saira@ubuntu:~$ su - Scooby
Password:
su: warning: cannot change directory to /home/Scooby: No such file or directory
$
```

5. Show that Scooby’s home directory does not exist yet and what /etc/passwd says:

exit

cat /etc/passwd

```
$ exit
saira@ubunto:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uuidd:x:104:105::/run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
saira:x:1000:1000:Saira:/home/saira:/bin/bash
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
rtkit:x:110:110:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
cups-pk-helper:x:111:111:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
lightdm:x:112:115:Light Display Manager:/var/lib/lightdm:/bin/false
whoopsie:x:113:117::/nonexistent:/bin/false
avahi:x:114:118:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
pulse:x:115:119:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
saned:x:116:122::/var/lib/saned:/usr/sbin/nologin
cups-browsed:x:117:111::/nonexistent:/usr/sbin/nologin
colord:x:118:123:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
xrdp:x:119:124::/run/xrdp:/usr/sbin/nologin
tom:x:1001:1004:Tom cruise,200,32455-432,12345-543,n/a:/home/tom:/bin/bash
Scooby:x:1002:1005::/home/Scooby:/bin/sh
saira@ubunto:~$ _
```

ls -ld /home/Scooby

```
saira@ubunto:~$ ls -ld /home/Scooby
ls: cannot access '/home/Scooby': No such file or directory
saira@ubunto:~$
```

6. Manually create Scooby’s home directory and set proper ownership and permissions:

sudo mkdir -p /home/Scooby

sudo chown Scooby:Scooby /home/Scooby

sudo chmod 750 /home/Scooby

ls -ld /home/Scooby

```
saira@ubunto:~$ sudo mkdir -p /home/Scooby
saira@ubunto:~$ sudo chown Scooby:Scooby /home/Scooby
saira@ubunto:~$ sudo chmod 750 /home/Scooby
saira@ubunto:~$ ls -ld /home/Scooby
drwxr-x--- 2 Scooby Scooby 4096 Nov  6 07:24 /home/Scooby
saira@ubunto:~$ _
```

7. Log in as Scooby again and verify you land in the correct home directory:

su - Scooby

pwd

ls -la

```
saira@ubunto:~$ su - Scooby
Password:
$ pwd
/home/Scooby
$ ls -la
total 8
drwxr-x--- 2 Scooby Scooby 4096 Nov  6 07:24 .
drwxr-xr-x 5 root   root   4096 Nov  6 07:24 ..
$ _
```

8. Verify users in system files and observe shell of Scooby**:**

exit

cat /etc/passwd

```
$ exit
saira@ubunto:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uuidd:x:104:105::/run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
saira:x:1000:1000:Saira:/home/saira:/bin/bash
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
rtkit:x:110:110:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
cups-pk-helper:x:111:111:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
lightdm:x:112:115:Light Display Manager:/var/lib/lightdm:/bin/false
whoopsie:x:113:117::/nonexistent:/bin/false
avahi:x:114:118:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
pulse:x:115:119:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
saned:x:116:122::/var/lib/saned:/usr/sbin/nologin
cups-browsed:x:117:111::/nonexistent:/usr/sbin/nologin
colord:x:118:123:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
xrdp:x:119:124::/run/xrdp:/usr/sbin/nologin
tom:x:1001:1004:Tom cruise,200,32455-432,12345-543,n/a:/home/tom:/bin/bash
Scooby:x:1002:1005::/home/Scooby:/bin/sh
saira@ubunto:~$ _
```

9. Change the shell from /bin/sh to /bin/bash

sudo usermod -s /bin/bash Scooby

su – Scooby

```
saira@ubunto:~$ sudo usermod -s /bin/bash Scooby
saira@ubunto:~$ su - Scooby
Password:
Scooby@ubunto:~$ _
```

10. Create groups:

sudo addgroup jolly

sudo groupadd anime

```
saira@ubunto:~$ sudo addgroup jolly
info: Selecting GID from range 1000 to 59999 ...
info: Adding group `jolly' (GID 1006) ...
saira@ubunto:~$ sudo groupadd anime
saira@ubunto:~$
```

10. Verify groups:

cat /etc/group

```
tss:x:108:
landscape:x:109:
fwupd-refresh:x:989:
saira:x:1000:
rtkit:x:110:
lpadmin:x:111:
netdev:x:112:
bluetooth:x:113:
ssl-cert:x:114:
lightdm:x:115:
nopasswdlogin:x:116:
whoopsie:x:117:
avahi:x:118:
pulse:x:119:
pulse-access:x:120:
scanner:x:121:saned
saned:x:122:
colord:x:123:
xrdp:x:124:
tom:x:1001:tom
developer:x:1002:
devops:x:1003:
designer:x:1004:
Scooby:x:1005:
jolly:x:1006:
anime:x:1007:
saira@ubunto:~$
```

11. Delete groups and users:

sudo delgroup jolly

sudo groupdel anime

cat /etc/group

```
utmp:x:43:
video:x:44:
sasl:x:45:
plugdev:x:46:saira
staff:x:50:
games:x:60:
users:x:100:
nogroup:x:65534:
systemd-journal:x:999:
systemd-network:x:998:
systemd-timesync:x:997:
input:x:996:
sgx:x:995:
kvm:x:994:
render:x:993:
lxd:x:101:saira
messagebus:x:102:
systemd-resolve:x:992:
_ssh:x:103:
polkitd:x:991:
crontab:x:990:
syslog:x:104:
uuidd:x:105:
rdma:x:106:
tcpdump:x:107:
tss:x:108:
landscape:x:109:
fwupd-refresh:x:989:
saira:x:1000:
rtkit:x:110:
lpadmin:x:111:
netdev:x:112:
bluetooth:x:113:
ssl-cert:x:114:
lightdm:x:115:
nopasswdlogin:x:116:
whoopsie:x:117:
avahi:x:118:
pulse:x:119:
pulse-access:x:120:
scanner:x:121:saned
saned:x:122:
colord:x:123:
xrdp:x:124:
tom:x:1001:tom
developer:x:1002:
devops:x:1003:
designer:x:1004:
Scooby:x:1005:
saira@ubunto:~$ _
```

sudo deluser --remove-home Jerry

sudo userdel -r Scooby

cat /etc/passwd

```
saira@ubunto:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uuidd:x:104:105::/run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
saira:x:1000:1000:Saira:/home/saira:/bin/bash
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
rtkit:x:110:110:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
cups-pk-helper:x:111:111:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
lightdm:x:112:115:Light Display Manager:/var/lib/lightdm:/bin/false
whoopsie:x:113:117::/nonexistent:/bin/false
avahi:x:114:118:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
pulse:x:115:119:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
saned:x:116:122::/var/lib/saned:/usr/sbin/nologin
cups-browsed:x:117:111::/nonexistent:/usr/sbin/nologin
colord:x:118:123:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
xrdp:x:119:124::/run/xrdp:/usr/sbin/nologin
tom:x:1001:1004:Tom cruise,200,32455-432,12345-543,n/a:/home/tom:/bin/bash
saira@ubunto:~$ _
```

Task 5 – Create user Student; create files; set owner/group; identify file types

1. Create Student:

sudo adduser Student

```
saira@ubunto:~$ sudo adduser student

info: Adding user `student' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `student' (1005) ...
info: Adding new user `student' (1005) with group `student (1005)' ...
info: Creating home directory `/home/student' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for student
Enter the new value, or press ENTER for the default
        Full Name []: student 1
        Room Number []: 22
        Work Phone []: 12345
        Home Phone []: 23456
        Other []: n/a
Is the information correct? [Y/n] y
info: Adding new user `student' to supplemental / extra groups `users' ...
info: Adding user `student' to group `users' ...
saira@ubunto:~$ _
```

2. Switch to Student and create files:

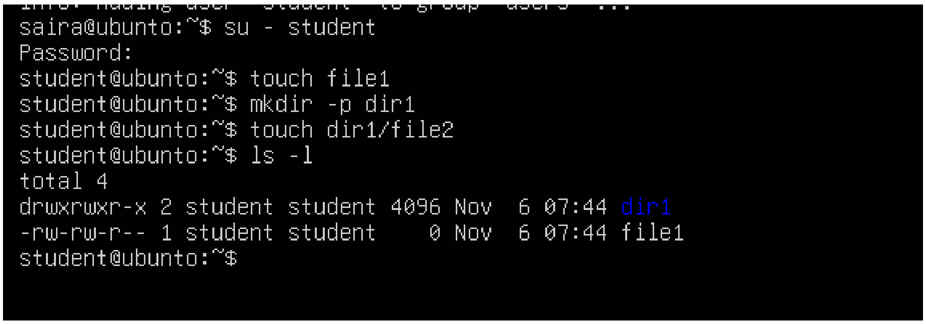
su - Student

touch file1

mkdir -p dir1

touch dir1/file2

ls -l

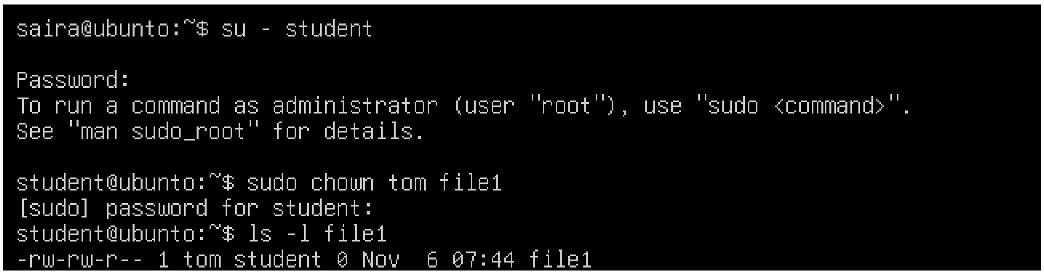


```
saira@ubunto:~$ su - student
Password:
student@ubunto:~$ touch file1
student@ubunto:~$ mkdir -p dir1
student@ubunto:~$ touch dir1/file2
student@ubunto:~$ ls -l
total 4
drwxrwxr-x 2 student student 4096 Nov  6 07:44 dir1
-rw-rw-r-- 1 student student    0 Nov  6 07:44 file1
student@ubunto:~$
```

3. Change owner then group for file1 (separate commands):

sudo chown tom file1

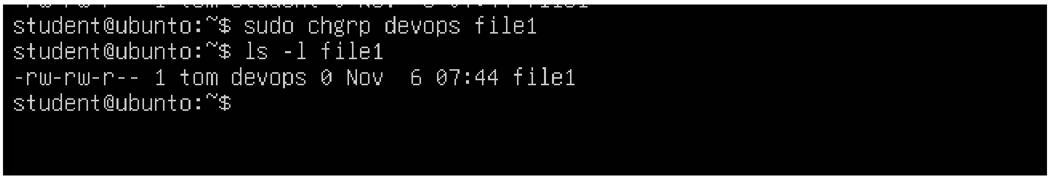
ls -l file1



```
saira@ubunto:~$ su - student

Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

student@ubunto:~$ sudo chown tom file1
[sudo] password for student:
student@ubunto:~$ ls -l file1
-rw-rw-r-- 1 tom student 0 Nov  6 07:44 file1
```

sudo chgrp devops file1

ls -l file1



```
student@ubunto:~$ sudo chgrp devops file1
student@ubunto:~$ ls -l file1
-rw-rw-r-- 1 tom devops 0 Nov  6 07:44 file1
student@ubunto:~$
```

4. Identify files/directories and show /dev/null:

ls -l

ls -l dir1

ls -l /dev/null

file file1 dir1 /dev/null

```
student@ubunto:~$ ls -l
total 4
drwxrwxr-x 2 student student 4096 Nov  6 07:44 dir1
-rw-rw-r-- 1 tom     devops     0 Nov  6 07:44 file1
student@ubunto:~$ ls -l dir1
total 0
-rw-rw-r-- 1 student student 0 Nov  6 07:44 file2
student@ubunto:~$ ls -l /dev/null
crw-rw-rw- 1 root root 1, 3 Nov  6 06:58 /dev/null
student@ubunto:~$ file file1 dir1 /dev/null
file1:     empty
dir1:      directory
/dev/null: character special (1/3)
student@ubunto:~$ _
```

5. Exit Student:

exit

```
saira@ubunto:~$ _
```

Task 6 – Change permissions using symbolic mode

1. Ensure Student and file present:

su - Student
cd ~
ls -l file1

```
saira@ubunto:~$ sudo chown student:student /home/student/file1


saira@ubunto:~$ ls -l /home/student/file1

ls: cannot access '/home/student/file1': Permission denied
saira@ubunto:~$
saira@ubunto:~$ sudo chown student:student /home/student/file1


saira@ubunto:~$ sudo ls -l /home/student/file1


-rw-rw-r-- 1 student student 0 Nov  6 07:44 /home/student/file1
saira@ubunto:~$ su - student
Password:
student@ubunto:~$ cd ~
student@ubunto:~$ ls -l file1
-rw-rw-r-- 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$
```

2. Remove all permissions:

chmod -rwx file1

ls -l file1

```
student@ubunto:~$ chmod -rwx file1
student@ubunto:~$ ls -l file1
---------- 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$
```

3. Add read to all:

chmod +r file1

ls -l file1

```
student@ubunto:~$ chmod +r file1
student@ubunto:~$ ls -l file1
-r--r--r-- 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$
```

4. Add execute to user:

chmod u+x file1

ls -l file1

```
student@ubunto:~$ chmod u+x file1
student@ubunto:~$ ls -l file1
-r-xr--r-- 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$ _
```

5. Add write to user and group:

chmod ug+w file1

ls -l file1

```
student@ubunto:~$ chmod ug+w file1
student@ubunto:~$ ls -l file1
-rwxrw-r-- 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$
```

6. Remove all permissions (explicit):

chmod ugo-rwx file1

ls -l file1

```
student@ubunto:~$ chmod ugo-rwx file1
student@ubunto:~$ ls -l file1
---------- 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$
```

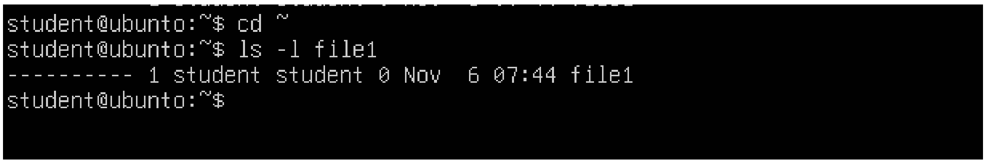
Task 7 – Change permissions using "set" symbolic form (u= g= o=)

Ensure you are Student:

su - Student

cd ~

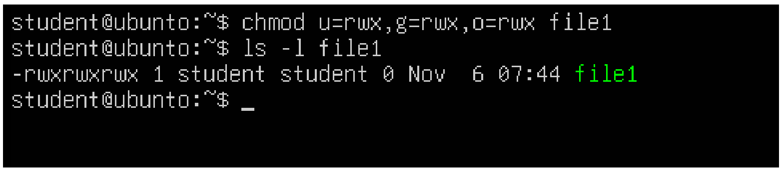
ls -l file1



1. Set all to rwx:

chmod u=rwx,g=rwx,o=rwx file1

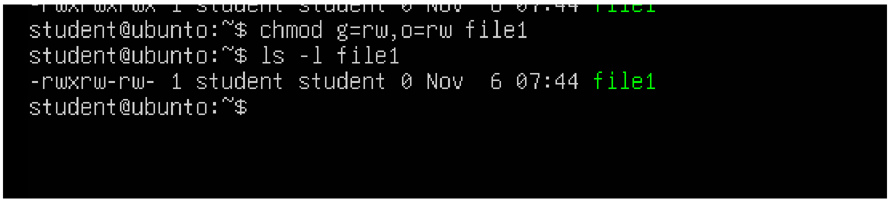
ls -l file1



2. Remove execute from group and others:

chmod g=rw,o=rw file1

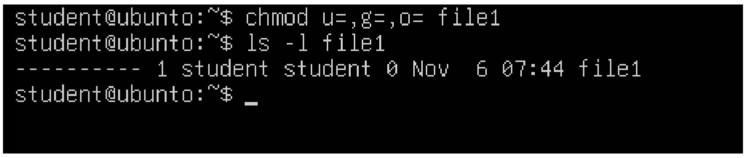
ls -l file1



3. Remove all permissions:

chmod u=,g=,o= file1

ls -l file1



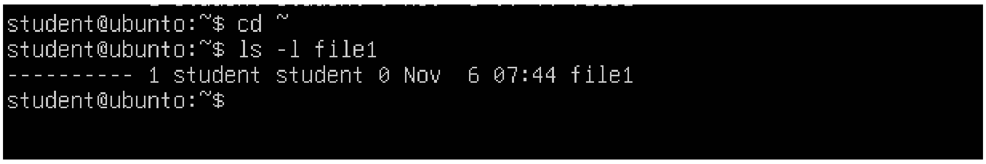
Task 8 – Change permissions using numeric (octal) mode

Ensure you are Student:

su - Student

cd ~

ls -l file1



Run each command and capture screenshot after each ls:

1.

chmod 777 file1

ls -l file1

```
student@ubunto:~$ chmod 777 file1
student@ubunto:~$ ls -l file1
-rwxrwxrwx 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$ _
```

2. **chmod 700 file1**

ls -l file1

```
student@ubunto:~$ chmod 700 file1
student@ubunto:~$ ls -l file1
-rwx------ 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$ _
```

3. **chmod 744 file1**

ls -l file1

```
student@ubunto:~$ chmod 744 file1
student@ubunto:~$ ls -l file1
-rwxr--r-- 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$
```

4. **chmod 640 file1**

ls -l file1

```
student@ubunto:~$ chmod 640 file1
student@ubunto:~$ ls -l file1
-rw-r----- 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$
```

5. **chmod 664 file1**

ls -l file1

```
student@ubunto:~$ chmod 664 file1
student@ubunto:~$ ls -l file1
-rw-rw-r-- 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$
```

6. **chmod 775 file1**

ls -l file1

```
student@ubunto:~$ chmod 775 file1
student@ubunto:~$ ls -l file1
-rwxrwxr-x 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$ _
```

7. **chmod 750 file1**

ls -l file1

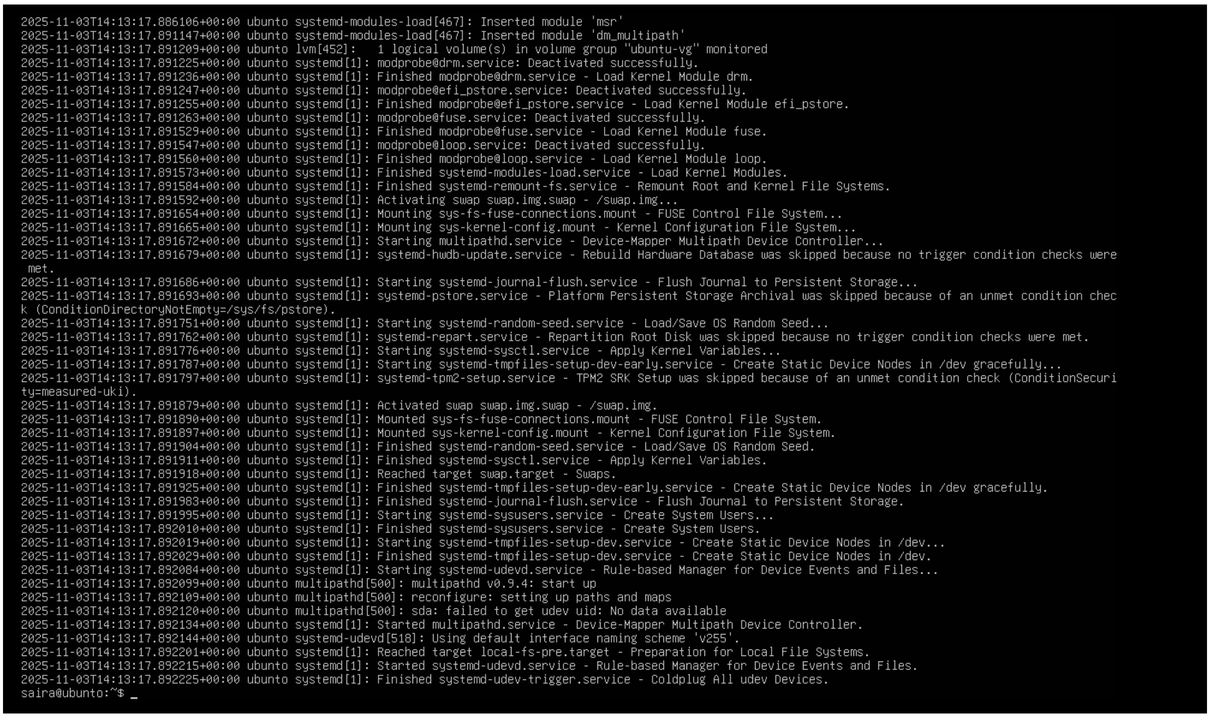
```
student@ubunto:~$ chmod 750 file1
student@ubunto:~$ ls -l file1
-rwxr-x--- 1 student student 0 Nov  6 07:44 file1
student@ubunto:~$
```

Task 9 – Practice pipes, pagers, grep, and redirects with /var/log/syslog

1. less:

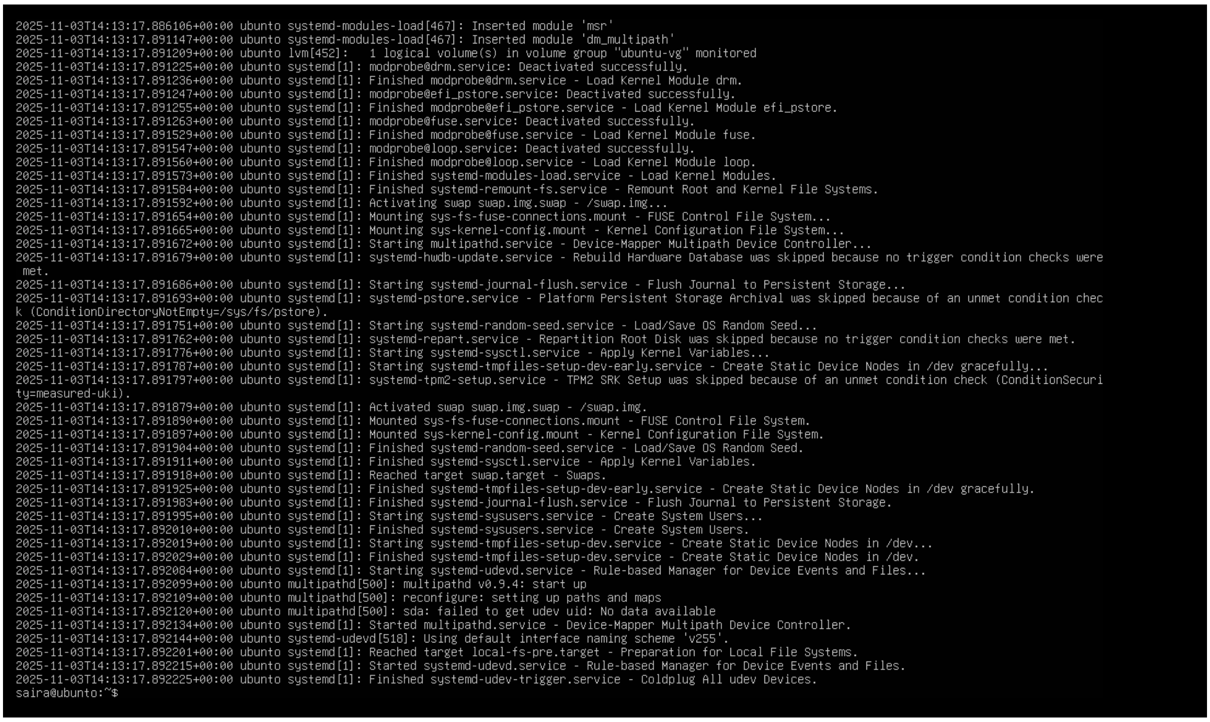
sudo cat /var/log/syslog | less

 # quit q



2. more:

sudo cat /var/log/syslog | more



3. grep failures/errors:

sudo grep -E 'fail|error' /var/log/syslog | head

```
saira@ubunto:~$ sudo grep -E 'fail|error' /var/log/syslog | head
2025-11-03T14:13:17.892120+00:00 ubunto multipathd[500]: sda: failed to get udev uid: No data available
2025-11-03T14:13:17.892330+00:00 ubunto multipath: sda: failed to get sysfs uid: No such file or directory
2025-11-03T14:13:17.892340+00:00 ubunto multipath: sda: failed to get sgio uid: No such file or directory
2025-11-03T14:13:17.892391+00:00 ubunto multipathd[500]: sda: failed to get udev uid: No data available
2025-11-03T14:13:17.905911+00:00 ubunto multipathd[500]: sda: failed to get path uid
2025-11-03T14:13:17.905956+00:00 ubunto multipathd[500]: uevent trigger error
2025-11-03T14:13:17.930779+00:00 ubunto multipath: sda: failed to get sysfs uid: No such file or directory
2025-11-03T14:13:17.930841+00:00 ubunto multipath: sda: failed to get sgio uid: No such file or directory
2025-11-03T14:13:17.930854+00:00 ubunto multipathd[500]: sda: failed to get udev uid: No data available
2025-11-03T14:13:17.930864+00:00 ubunto multipathd[500]: sda: failed to get path uid
grep: /var/log/syslog: binary file matches
saira@ubunto:~$
```

4. redirect:

sudo grep -i systemd /var/log/syslog > ~/syslog_systemd.txt

```
grep: /var/log/syslog: binary file matches
saira@ubunto:~$ sudo grep -i systemd /var/log/syslog > ~/syslog_systemd.txt
grep: /var/log/syslog: binary file matches
saira@ubunto:~$ _
```

5. append:

sudo grep -i network /var/log/syslog >> ~/syslog_systemd.txt

```
saira@ubunto:~$ sudo grep -i network /var/log/syslog >> ~/syslog_systemd.txt
grep: /var/log/syslog: binary file matches
saira@ubunto:~$ _
```

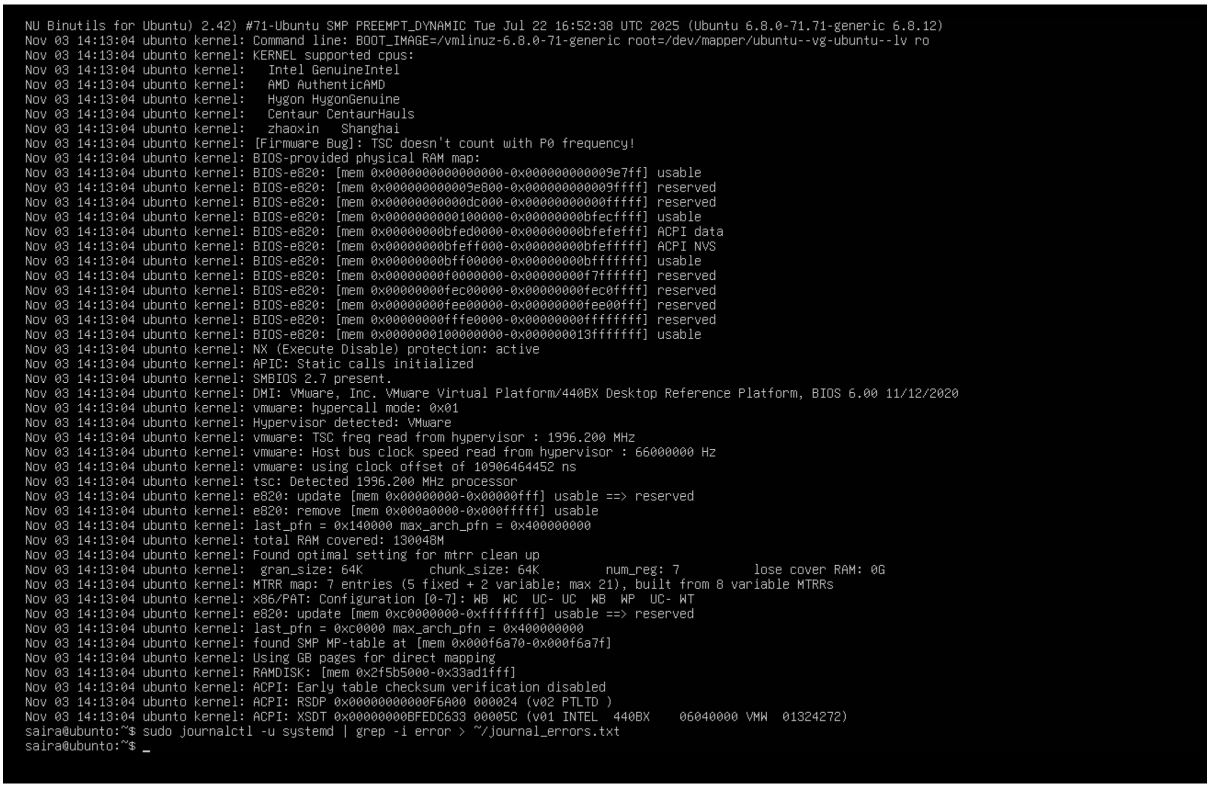
cat ~/syslog_systemd.txt

```
2025-11-04T11:35:01.605918+00:00 ubunto NetworkManager[867]: <info>  [1762256101.6036] monitoring ifupdown state file '/run/network/ifstate'.
2025-11-04T11:35:01.609794+00:00 ubunto dbus-daemon[842]: [system] Activating via systemd: service name='org.freedesktop.hostname1' unit='dbus-org.freedesktop.h
ostname1.service' requested by ':1.9' (uid=0 pid=867 comm="/usr/sbin/NetworkManager --no-daemon" label="unconfined")
2025-11-04T11:35:01.963918+00:00 ubunto NetworkManager[867]: <info>  [1762256101.9637] hostname: hostname: using hostnamed
2025-11-04T11:35:01.964004+00:00 ubunto NetworkManager[867]: <info>  [1762256101.9638] hostname: static hostname changed from (none) to "ubunto"
2025-11-04T11:35:01.967468+00:00 ubunto NetworkManager[867]: <info>  [1762256101.9663] dns-mgr: init: dns=systemd-resolved rc-manager=unmanaged (auto), plugin=s
ystemd-resolved
2025-11-04T11:35:01.967541+00:00 ubunto NetworkManager[867]: <info>  [1762256101.9673] manager[0x615ad9fb9a40]: rfkill: Wi-Fi hardware radio set enabled
2025-11-04T11:35:01.967585+00:00 ubunto NetworkManager[867]: <info>  [1762256101.9674] manager[0x615ad9fb9a40]: rfkill: WWAN hardware radio set enabled
2025-11-04T11:35:02.005486+00:00 ubunto NetworkManager[867]: <info>  [1762256102.0051] Loaded device plugin: NMBluezManager (/usr/lib/x86_64-linux-gnu/NetworkMa
nager/1.46.0/libnm-device-plugin-bluetooth.so)
2025-11-04T11:35:02.012068+00:00 ubunto NetworkManager[867]: <info>  [1762256102.0116] Loaded device plugin: NMWwanFactory (/usr/lib/x86_64-linux-gnu/NetworkMan
ager/1.46.0/libnm-device-plugin-wwan.so)
2025-11-04T11:35:02.029091+00:00 ubunto NetworkManager[867]: <info>  [1762256102.0277] Loaded device plugin: NMTeamFactory (/usr/lib/x86_64-linux-gnu/NetworkMan
ager/1.46.0/libnm-device-plugin-team.so)
2025-11-04T11:35:02.048207+00:00 ubunto NetworkManager[867]: <info>  [1762256102.0456] Loaded device plugin: NMWifiFactory (/usr/lib/x86_64-linux-gnu/NetworkMan
ager/1.46.0/libnm-device-plugin-wifi.so)
2025-11-04T11:35:02.058734+00:00 ubunto NetworkManager[867]: <info>  [1762256102.0549] Loaded device plugin: NMAtmManager (/usr/lib/x86_64-linux-gnu/NetworkMana
ger/1.46.0/libnm-device-plugin-adsl.so)
2025-11-04T11:35:02.058819+00:00 ubunto NetworkManager[867]: <info>  [1762256102.0555] manager: rfkill: Wi-Fi enabled by radio killswitch; enabled by state file
2025-11-04T11:35:02.058858+00:00 ubunto NetworkManager[867]: <info>  [1762256102.0555] manager: rfkill: WWAN enabled by radio killswitch; enabled by state file
2025-11-04T11:35:02.058889+00:00 ubunto NetworkManager[867]: <info>  [1762256102.0556] manager: Networking is enabled by state file
2025-11-04T11:35:02.064569+00:00 ubunto dbus-daemon[842]: [system] Activating via systemd: service name='org.freedesktop.nm_dispatcher' unit='dbus-org.freedeskt
op.nm-dispatcher.service' requested by ':1.9' (uid=0 pid=867 comm="/usr/sbin/NetworkManager --no-daemon" label="unconfined")
2025-11-04T11:35:02.077628+00:00 ubunto NetworkManager[867]: <info>  [1762256102.0746] settings: Loaded settings plugin: ifupdown ("/usr/lib/x86_64-linux-gnu/Ne
tworkManager/1.46.0/libnm-settings-plugin-ifupdown.so")
2025-11-04T11:35:02.082961+00:00 ubunto NetworkManager[867]: <info>  [1762256102.0763] settings: Loaded settings plugin: keyfile (internal)
2025-11-04T11:35:02.083058+00:00 ubunto NetworkManager[867]: <info>  [1762256102.0763] ifupdown: management mode: unmanaged
2025-11-04T11:35:02.087021+00:00 ubunto NetworkManager[867]: <info>  [1762256102.0767] ifupdown: interfaces file /etc/network/interfaces doesn't exist
2025-11-04T11:35:02.087406+00:00 ubunto systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service...
2025-11-04T11:35:02.224789+00:00 ubunto systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
2025-11-04T11:35:03.436317+00:00 ubunto systemd[1]: Reloading requested from client PID 1031 ('systemctl') (unit NetworkManager.service)...
2025-11-04T11:35:04.145288+00:00 ubunto NetworkManager[867]: <info>  [1762256104.1428] dhcp: init: Using DHCP client 'internal'
2025-11-04T11:35:04.145394+00:00 ubunto NetworkManager[867]: <info>  [1762256104.1431] manager: (lo): new Loopback device (/org/freedesktop/NetworkManager/Devic
es/1)
2025-11-04T11:35:04.145442+00:00 ubunto NetworkManager[867]: <info>  [1762256104.1439] device (ens33): carrier: link connected
2025-11-04T11:35:04.145562+00:00 ubunto NetworkManager[867]: <info>  [1762256104.1444] manager: (ens33): new Ethernet device (/org/freedesktop/NetworkManager/De
vices/2)
2025-11-04T11:35:04.150846+00:00 ubunto NetworkManager[867]: <info>  [1762256104.1464] failed to open /run/network/ifstate
2025-11-04T11:35:04.150971+00:00 ubunto NetworkManager[867]: <info>  [1762256104.1470] bus-manager: acquired D-Bus service "org.freedesktop.NetworkManager"
2025-11-04T11:35:04.151015+00:00 ubunto NetworkManager[867]: <info>  [1762256104.1478] manager: startup complete
2025-11-04T11:35:04.151057+00:00 ubunto systemd[1]: Started NetworkManager.service - Network Manager.
2025-11-04T11:35:04.154866+00:00 ubunto systemd[1]: Reached target network.target - Network.
2025-11-04T11:35:04.166582+00:00 ubunto NetworkManager[867]: <info>  [1762256104.1664] modem-manager: ModemManager available
2025-11-04T11:35:04.172271+00:00 ubunto systemd[1]: Starting NetworkManager-wait-online.service - Network Manager Wait Online...
2025-11-04T11:35:04.273199+00:00 ubunto systemd[1]: Finished NetworkManager-wait-online.service - Network Manager Wait Online.
2025-11-04T11:35:04.291179+00:00 ubunto systemd[1]: Reached target network-online.target - Network is Online.
2025-11-04T11:35:12.238406+00:00 ubunto systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
2025-11-04T11:35:22.459551+00:00 ubunto systemd[1521]: Listening on dirmngr.socket - GnuPG network certificate management daemon.
saira@ubunto:~$
```

Alternative (journalctl) if needed:

sudo journalctl | less

sudo journalctl -u systemd | grep -i error > ~/journal_errors.txt



```
NU Binutils for Ubuntu) 2.42) #71-Ubuntu SMP PREEMPT_DYNAMIC Tue Jul 22 16:52:38 UTC 2025 (Ubuntu 6.8.0-71.71-generic 6.8.12)
Nov 03 14:13:04 ubunto kernel: Command line: BOOT_IMAGE=/vmlinuz-6.8.0-71-generic root=/dev/mapper/ubuntu--vg-ubuntu--lv ro
Nov 03 14:13:04 ubunto kernel: KERNEL supported cpus:
Nov 03 14:13:04 ubunto kernel:   Intel GenuineIntel
Nov 03 14:13:04 ubunto kernel:   AMD AuthenticAMD
Nov 03 14:13:04 ubunto kernel:   Hygon HygonGenuine
Nov 03 14:13:04 ubunto kernel:   Centaur CentaurHauls
Nov 03 14:13:04 ubunto kernel:   zhaoxin   Shanghai
Nov 03 14:13:04 ubunto kernel: [Firmware Bug]: TSC doesn't count with P0 frequency!
Nov 03 14:13:04 ubunto kernel: BIOS-provided physical RAM map:
Nov 03 14:13:04 ubunto kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009e7ff] usable
Nov 03 14:13:04 ubunto kernel: BIOS-e820: [mem 0x000000000009e800-0x000000000009ffff] reserved
Nov 03 14:13:04 ubunto kernel: BIOS-e820: [mem 0x00000000000dc000-0x00000000000fffff] reserved
Nov 03 14:13:04 ubunto kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000bfecffff] usable
Nov 03 14:13:04 ubunto kernel: BIOS-e820: [mem 0x00000000bfed0000-0x00000000bfefefff] ACPI data
Nov 03 14:13:04 ubunto kernel: BIOS-e820: [mem 0x00000000bfeff000-0x00000000bfefffff] ACPI NVS
Nov 03 14:13:04 ubunto kernel: BIOS-e820: [mem 0x00000000bff00000-0x00000000bfffffff] usable
Nov 03 14:13:04 ubunto kernel: BIOS-e820: [mem 0x00000000f0000000-0x00000000f7ffffff] reserved
Nov 03 14:13:04 ubunto kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec0ffff] reserved
Nov 03 14:13:04 ubunto kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
Nov 03 14:13:04 ubunto kernel: BIOS-e820: [mem 0x00000000fffe0000-0x00000000ffffffff] reserved
Nov 03 14:13:04 ubunto kernel: BIOS-e820: [mem 0x0000000100000000-0x000000013fffffff] usable
Nov 03 14:13:04 ubunto kernel: NX (Execute Disable) protection: active
Nov 03 14:13:04 ubunto kernel: APIC: Static calls initialized
Nov 03 14:13:04 ubunto kernel: SMBIOS 2.7 present.
Nov 03 14:13:04 ubunto kernel: DMI: VMware, Inc. VMware Virtual Platform/440BX Desktop Reference Platform, BIOS 6.00 11/12/2020
Nov 03 14:13:04 ubunto kernel: vmware: hypercall mode: 0x01
Nov 03 14:13:04 ubunto kernel: Hypervisor detected: VMware
Nov 03 14:13:04 ubunto kernel: vmware: TSC freq read from hypervisor : 1996.200 MHz
Nov 03 14:13:04 ubunto kernel: vmware: Host bus clock speed read from hypervisor : 66000000 Hz
Nov 03 14:13:04 ubunto kernel: vmware: using clock offset of 10906464452 ns
Nov 03 14:13:04 ubunto kernel: tsc: Detected 1996.200 MHz processor
Nov 03 14:13:04 ubunto kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Nov 03 14:13:04 ubunto kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
Nov 03 14:13:04 ubunto kernel: last_pfn = 0x140000 max_arch_pfn = 0x400000000
Nov 03 14:13:04 ubunto kernel: total RAM covered: 130048M
Nov 03 14:13:04 ubunto kernel: Found optimal setting for mtrr clean up
Nov 03 14:13:04 ubunto kernel:  gran_size: 64K         chunk_size: 64K         num_reg: 7          lose cover RAM: 0G
Nov 03 14:13:04 ubunto kernel: MTRR map: 7 entries (5 fixed + 2 variable; max 21), built from 8 variable MTRRs
Nov 03 14:13:04 ubunto kernel: x86/PAT: Configuration [0-7]: WB  WC  UC- UC  WB  WP  UC- WT
Nov 03 14:13:04 ubunto kernel: e820: update [mem 0xc0000000-0xffffffff] usable ==> reserved
Nov 03 14:13:04 ubunto kernel: last_pfn = 0xc0000 max_arch_pfn = 0x400000000
Nov 03 14:13:04 ubunto kernel: found SMP MP-table at [mem 0x000f6a70-0x000f6a7f]
Nov 03 14:13:04 ubunto kernel: Using GB pages for direct mapping
Nov 03 14:13:04 ubunto kernel: RAMDISK: [mem 0x2f5b5000-0x33ad1fff]
Nov 03 14:13:04 ubunto kernel: ACPI: Early table checksum verification disabled
Nov 03 14:13:04 ubunto kernel: ACPI: RSDP 0x00000000000F6A00 000024 (v02 PTLTD )
Nov 03 14:13:04 ubunto kernel: ACPI: XSDT 0x00000000BFEDC633 00005C (v01 INTEL  440BX    06040000 VMW  01324272)
saira@ubunto:~$ sudo journalctl -u systemd | grep -i error > ~/journal_errors.txt
saira@ubunto:~$ _
```

Task 10 – Script setup.sh – variables, command substitution, file/dir checks, permissions (use vim)

Start in your Student home directory (recommended).

1. Include bash shebang

- Code to add (enter in vim as the first line of the file):

#!/bin/bash

- Steps:
  - i. **vim setup.sh → add the shebang line → save and quit**
  - ii. **chmod +x setup.sh**
  - iii. **./setup.sh**

```
#!/bin/bash
~
:wq
```

```
"setup.sh" [New] 1L, 12B written
student@ubunto:~$ chmod +x setup.sh
student@ubunto:~$
student@ubunto:~$ ./setup.sh
student@ubunto:~$
student@ubunto:~$
```

Step B2 — Append var1

1. Open setup.sh again:

vim setup.sh

2. Press i and go to new line under the shebang, paste:

 # Define and show var1

var1="Hello from Lab 6"

echo "var1: $var1"

```
#!/bin/bash
# Define and show var1
var1="Hello from Lab 6"
echo "var1: $var1"
~
```

3. Save and quit:

ESC

:wq

4. Run the script:

./setup.sh

```
"setup.sh" 4L, 78B written
student@ubunto:~$ ./setup.sh
var1: Hello from Lab 6
student@ubunto:~$
```

Step B3 — Append ls -l variable

vim setup.sh

Insert below previous code:

 # Save ls -l to variable and display

allFiles="$(ls -l)"
echo "allFiles (ls -l):"
echo "$allFiles"

```
#!/bin/bash
# Define and show var1
var1="Hello from Lab 6"
echo "var1: $var1"
# Save ls -l to variable and display
 allFiles="$(ls -l)"
echo "allFiles (ls -l):"

echo "$allFiles"

~
~
```

Run:

./setup.sh

```
student@ubunto:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 332
drwxrwxr-x 2 student student   4096 Nov  6 07:44 dir1
-rwxr-x--- 1 student student      0 Nov  6 07:44 file1
-rw-rw-r-- 1 student student      0 Nov  6 08:22 journal_errors.txt
-rwxrwxr-x 1 student student    180 Nov  6 08:43 setup.sh
-rw-rw-r-- 1 student student 328889 Nov  6 08:21 syslog_systemd.txt
student@ubunto:~$
```

Step B4 — Directory check

vim setup.sh

Add:

 # Directory check
if [-d "dir1"]; then
  echo "Directory dir1 exists."
else
  echo "Directory dir1 does not exist. Creating..."
  mkdir -p "dir1"
  echo "Directory dir1 created."
Fi

```
#!/bin/bash
# Define and show var1
var1="Hello from Lab 6"
echo "var1: $var1"
# Save ls -l to variable and display
 allFiles="$(ls -l)"
echo "allFiles (ls -l):"

echo "$allFiles"
if [ -d "dir1" ]; then

          echo "Directory dir1 exists."

  else

            echo "Directory dir1 does not exist. Creating..."

              mkdir -p "dir1"

                  echo "Directory dir1 created."

fi_
~
```

Save & Run:

./setup.sh

```
~
"setup.sh" 22L, 369B written
student@ubunto:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 332
drwxrwxr-x 2 student student   4096 Nov  6 07:44 dir1
-rwxr-x--- 1 student student      0 Nov  6 07:44 file1
-rw-rw-r-- 1 student student      0 Nov  6 08:22 journal_errors.txt
-rwxrwxr-x 1 student student    369 Nov  6 08:45 setup.sh
-rw-rw-r-- 1 student student 328889 Nov  6 08:21 syslog_systemd.txt
Directory dir1 exists.
student@ubunto:~$ _
```

Step B5 — File check

vim setup.sh

Add:

 # File check
if [-f "dir1/file2"]; then
  echo "file2 already exists."
else
  echo "file2 does not exist. Creating..."

touch "dir1/file2"
chmod a-rwx "dir1/file2"
echo "file2 created."
Fi

```
#!/bin/bash
# Define and show var1
var1="Hello from Lab 6"
echo "var1: $var1"
# Save ls -l to variable and display
 allFiles="$(ls -l)"
echo "allFiles (ls -l):"

echo "$allFiles"
 if [ -f "dir1/file2" ]; then

           echo "file2 already exists."

   else

             echo "file2 does not exist. Creating..."

               touch "dir1/file2"

                 chmod a-rwx "dir1/file2"

                   echo "file2 created."

 fi_
~
```

Save & Run:

./setup.sh

```
"setup.sh" 24L, 397B written
student@ubunto:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 332
drwxrwxr-x 2 student student   4096 Nov  6 07:44 dir1
-rwxr-x--- 1 student student      0 Nov  6 07:44 file1
-rw-rw-r-- 1 student student      0 Nov  6 08:22 journal_errors.txt
-rwxrwxr-x 1 student student    397 Nov  6 08:47 setup.sh
-rw-rw-r-- 1 student student 328889 Nov  6 08:21 syslog_systemd.txt
file2 already exists.
student@ubunto:~$ _
```

Step B6 — Permission checks

vim setup.sh

Add:

Permission checks for dir1/file2 (user permissions)
f="dir1/file2"
if [! -r "$f"]; then
echo "Read permission missing; granting to user..."
chmod u+r "$f"
fi

if [! -w "$f"]; then

```
  echo "Write permission missing; granting to user..."
  chmod u+w "$f"
fi

if [ ! -x "$f" ]; then
  echo "Execute permission missing; granting to user..."
  chmod u+x "$f"
fi

echo "Final permissions for $f:"
ls -l "$f"
```

```
#!/bin/bash
# Define and show var1
var1="Hello from Lab 6"
echo "var1: $var1"
# Save ls -l to variable and display
 allFiles="$(ls -l)"
echo "allFiles (ls -l):"

   _Permission checks for dir1/file2 (user permissions)

   f="dir1/file2"

   if [ ! -r "$f" ]; then

     echo "Read permission missing; granting to user..."

       chmod u+r "$f"

       fi


       if [ ! -w "$f" ]; then

         echo "Write permission missing; granting to user..."

           chmod u+w "$f"

            fi


           if [ ! -x "$f" ]; then

             echo "Execute permission missing; granting to user..."

               chmod u+x "$f"
                fi
              echo "Final permissions for $f:"

  ls -l "$f"
~
~
~
~
~
~
~
~
-- INSERT --
```

Save & Run:

./setup.sh

```
"setup.sh" 41L, 746B written
student@ubunto:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
Execute permission missing; granting to user...
Final permissions for dir1/file2:
-rwxrw-r-- 1 student student 0 Nov  6 07:44 dir1/file2
student@ubunto:~$
```

Task 11 – Script setup.sh – argument comparisons (eq, ne, gt, lt, ge, le) and string checks

STEP B0 — Create File and Define Variables

Open the file:

vim setup.sh

Insert exactly this inside**:**

#!/bin/bash

num=$1

str=$2

Save and exit:

:wq

Make executable:

chmod +x setup.sh

Run example:

./setup.sh 10 Student

STEP B1 — Add -eq (equal)

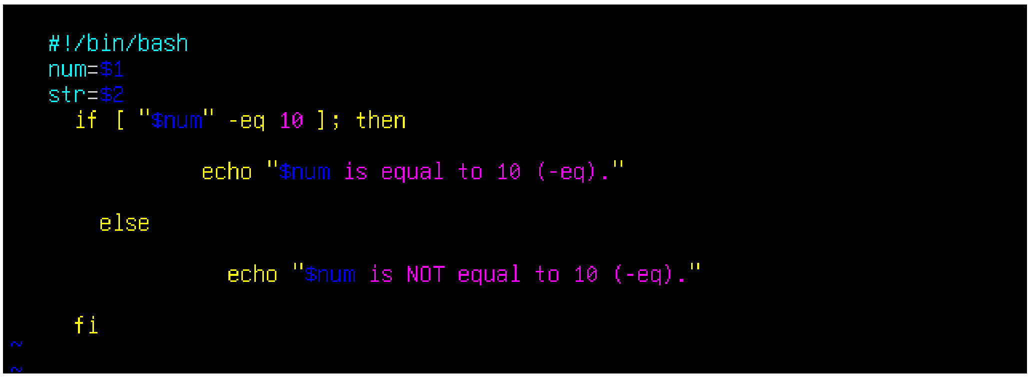
Open for edit:

vim setup.sh

Append this at the bottom:

```
if [ "$num" -eq 10 ]; then
  echo "$num is equal to 10 (-eq)."
else
  echo "$num is NOT equal to 10 (-eq)."
fi
```

Save & exit: :wq



Run:

./setup.sh 10 Student

./setup.sh 7 Student

```
"setup.sh" 12L, 181B written
student@ubunto:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
student@ubunto:~$ ./setup.sh 7 Student
7 is NOT equal to 10 (-eq).
student@ubunto:~$ _
```

STEP B2 — Add -ne (not equal)

Append:

if ["$num" -ne 10]; then

echo "$num is not equal to 10 (-ne)."

else

echo "$num is equal to 10 (-ne false)."

Fi

```
if [ "$num" -ne 10 ]; then
          echo "$num is not equal to 10 (-ne)."
  else
            echo "$num is equal to 10 (-ne false)."
fi
```

Run:

./setup.sh 7 Student

./setup.sh 10 Student

```
"setup.sh" 16L, 182B written
student@ubunto:~$ ./setup.sh 10 Student
10 is equal to 10 (-ne false).
student@ubunto:~$ ./setup.sh 7 Student
7 is not equal to 10 (-ne).
student@ubunto:~$ _
```

STEP B3 — Add -gt (greater than)

Append:

if ["$num" -gt 10]; then

echo "$num is greater than 10 (-gt)."

else

echo "$num is NOT greater than 10 (-gt)."

Fi

```
#!/bin/bash
num=$1
str=$2


        if [ "$num" -gt 10 ]; then
                  echo "$num is greater than 10 (-gt)."
          else
                    echo "$num is NOT greater than 10 (-gt)."
        fi
```

Run:

./setup.sh 12 Student

./setup.sh 9 Student

```
"setup.sh" 17L, 200B written
student@ubunto:~$ ./setup.sh 10 Student
10 is NOT greater than 10 (-gt).
student@ubunto:~$ ./setup.sh 7 Student
7 is NOT greater than 10 (-gt).
student@ubunto:~$
```

STEP B4 — Add -lt (less than)

Append:

if ["$num" -lt 10]; then

echo "$num is less than 10 (-lt)."

else

echo "$num is NOT less than 10 (-lt)."

Fi

```
#!/bin/bash
num=$1
str=$2


        if [ "$num" -lt 10 ]; then
                  echo "$num is less than 10 (-lt)."
          else
                    echo "$num is NOT less than 10 (-lt)."
        fi
```

Run:

./setup.sh 5 Student

./setup.sh 11 Student

```
"setup.sh" 16L, 188B written
student@ubunto:~$ ./setup.sh 10 Student
10 is NOT less than 10 (-lt).
student@ubunto:~$ ./setup.sh 7 Student
7 is less than 10 (-lt).
student@ubunto:~$
```

STEP B5 — Add -ge (≥)

Append:

if ["$num" -ge 10]; then

echo "$num is greater than or equal to 10 (-ge)."

else

echo "$num is NOT greater than or equal to 10 (-ge)."

Fi

```
#!/bin/bash
num=$1
str=$2

        if [ "$num" -ge 10 ]; then

                echo "$num is greater than or equal to 10 (-ge)."

         else

                 echo "$num is NOT greater than or equal to 10 (-ge)."

        fi
~
```

Run:

./setup.sh 10 Student

./setup.sh 8 Student

```
"setup.sh" 14L, 202B written
student@ubunto:~$ ./setup.sh 10 Student
10 is greater than or equal to 10 (-ge).
student@ubunto:~$ ./setup.sh 7 Student
7 is NOT greater than or equal to 10 (-ge).
student@ubunto:~$
```

STEP B6 — Add -le (≤)

Append:

if ["$num" -le 10]; then

echo "$num is less than or equal to 10 (-le)."

else

echo "$num is NOT less than or equal to 10 (-le)."

Fi

```
#!/bin/bash
num=$1
str=$2

 if [ "$num" -le 10 ]; then

          echo "$num is less than or equal to 10 (-le)."

   else

            echo "$num is NOT less than or equal to 10 (-le)."

 fi
```

Run:

./setup.sh 10 Student

./setup.sh 12 Student

```
~
"setup.sh" 15L, 205B written
student@ubunto:~$ ./setup.sh 10 Student
10 is less than or equal to 10 (-le).
student@ubunto:~$ ./setup.sh 7 Student
7 is less than or equal to 10 (-le).
student@ubunto:~$ _
```

STEP B7 — String Equality =

Append:

if ["$str" = "Student"]; then

echo "Second argument equals 'Student' (=)."

else

echo "Second argument does NOT equal 'Student' (=)."

Fi

```
#!/bin/bash
num=$1
str=$2

     if [ "$str" = "Student" ]; then

               echo "Second argument equals 'Student' ( = )."

       else

                 echo "Second argument does NOT equal 'Student' ( = )."

     fi
```

Run:

./setup.sh 10 Student

./setup.sh 10 Test

```
"setup.sh" 13L, 214B written
student@ubunto:~$ ./setup.sh 10 Student
Second argument equals 'Student' ( = ).
student@ubunto:~$ ./setup.sh 7 Student
Second argument equals 'Student' ( = ).
student@ubunto:~$ _
```

STEP B8 — String Inequality !=

Append:

if ["$str" != "Student"]; then

echo "Second argument is not equal to 'Student' (!=)."

else

echo "Second argument equals 'Student' (!= false)."

Fi

```
#!/bin/bash
num=$1
str=$2

     if [ "$str" != "Student" ]; then

       echo "Second argument is not equal to 'Student' ( != )."

else

echo "Second argument equals 'Student' ( != false)."

     fi
~
~
~
```

Run:

./setup.sh 10 Test

./setup.sh 10 Student

```
"setup.sh" 13L, 218B written
student@ubunto:~$ ./setup.sh 10 Student
Second argument equals 'Student' ( != false).
student@ubunto:~$ ./setup.sh 7 Student
Second argument equals 'Student' ( != false).
student@ubunto:~$ _
```

STEP B9 — Check if second argument is empty -z

Append:

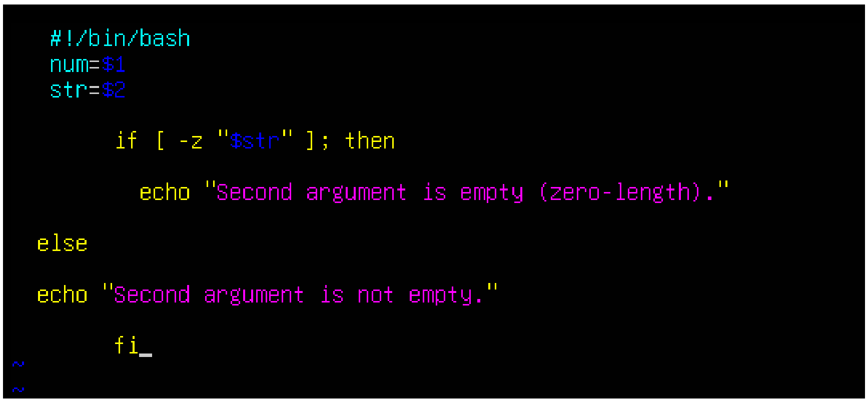
if [-z "$str"]; then

echo "Second argument is empty (zero-length)."

else

echo "Second argument is not empty."

Fi

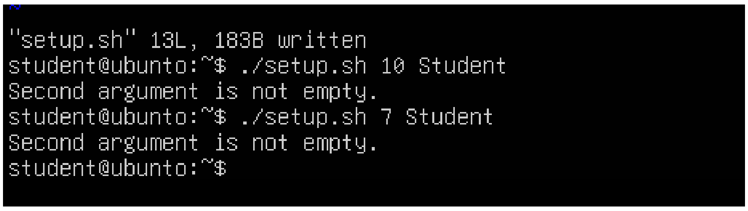


```
#!/bin/bash
num=$1
str=$2

     if [ -z "$str" ]; then

       echo "Second argument is empty (zero-length)."

else

echo "Second argument is not empty."

     fi_
~
~
```

Run:

./setup.sh 10

./setup.sh 10 Student



```
"setup.sh" 13L, 183B written
student@ubunto:~$ ./setup.sh 10 Student
Second argument is not empty.
student@ubunto:~$ ./setup.sh 7 Student
Second argument is not empty.
student@ubunto:~$
```

Combined:

```
                echo "$num is less than 10 (-lt)."
          else
                 echo "$num is NOT less than 10 (-lt)."
         fi

         if [ "$num" -ge 10 ]; then
                echo "$num is greater than or equal to 10 (-ge)."
          else
                 echo "$num is NOT greater than or equal to 10 (-ge)."
         fi

         if [ "$num" -le 10 ]; then
                echo "$num is less than or equal to 10 (-le)."
          else
                 echo "$num is NOT less than or equal to 10 (-le)."
         fi

         if [ "$str" = "Student" ]; then
                echo "Second argument equals 'Student' ( = )."
          else
                 echo "Second argument does NOT equal 'Student' ( = )."
         fi

         if [ "$str" != "Student" ]; then
                echo "Second argument is not equal _
-- INSERT --
```

```
saira@ubunto:~$ chmod +x setup.sh
saira@ubunto:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
./setup.sh: line 96: unexpected EOF while looking for matching `"'
saira@ubunto:~$
```

Task 12 – Script setup.sh – print all arguments with a for loop

1. Create the script with shebang and basic structure

- Open vim and overwrite setup.sh:

vim setup.sh

- Insert these lines (first step — shebang and a short comment):

#!/bin/bash

 # Script to demonstrate printing all user-entered arguments using $*

```
#!/bin/bash
#
## Script to demonstrate printing all user-entered arguments using $*_
```

- Save and quit (:wq)

2. Append the for loop using $* and print each argument

- Re-open setup.sh in vim and append the following lines:

 # Print all arguments using $*

echo "Printing all arguments using \$*:"

for arg in $*; do

echo "Argument: $arg"

done

```
#!/bin/bash
#
## Script to demonstrate printing all user-entered arguments using $*
    # Print all arguments using $*

     echo "Printing all arguments using \$*:"

     for arg in $*; do

       echo "Argument: $arg"

       done
```

- Save and quit (:wq)
- Make the script executable and run it with example arguments:

chmod +x setup.sh

./setup.sh one "two words" three

```
"setup.sh" 12L, 253B written
student@ubunto:~$ chmod +x setup.sh
student@ubunto:~$ ./setup.sh one "two words" three
Printing all arguments using $*:
Argument: one
Argument: two
Argument: words
Argument: three
student@ubunto:~$
```

Task 13 – Script setup.sh – while loop summation and functions

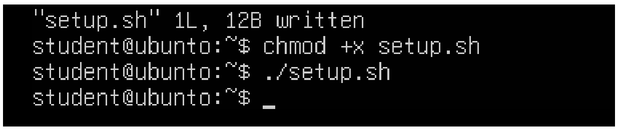
1. Add the shebang line

- Open vim and overwrite setup.sh with the shebang line:

```
#!/bin/bash
```

- Save and quit (:wq)
- Make executable and run (no output expected):

```
chmod +x setup.sh
./setup.sh
```



2. **Add the while-loop summation (interactive)**

- Re-open setup.sh in vim and append the while-loop:

```
# While-loop summation (interactive)
sum=0
while true; do
  read -p "Enter a number (or 'q' to quit): " input
  if [ "$input" = "q" ]; then
    break
  fi

  sum=$((sum + input))
  echo "Total Score: $sum"
done
echo "Final total: $sum"
```

```
#!/bin/bash
          sum=0

          while true; do

                    read -p "Enter a number (or 'q' to quit): " input

                      if [ "$input" = "q" ]; then

                                  break

                                    fi


                                      sum=$((sum + input))

                                        echo "Total Score: $sum"

                                done

                                echo "Final total: $sum"_
~
~
```

- Save and quit (:wq)
- Run the script and demonstrate a short session (example): enter 5, then 7, then q

./setup.sh

```
"setup.sh" 22L, 285B written
student@ubunto:~$ ./setup.sh
Enter a number (or 'q' to quit): 5
Total Score: 5
Enter a number (or 'q' to quit): 7
Total Score: 12
Enter a number (or 'q' to quit): 5
Total Score: 17
Enter a number (or 'q' to quit): q
Final total: 17
student@ubunto:~$ _
```

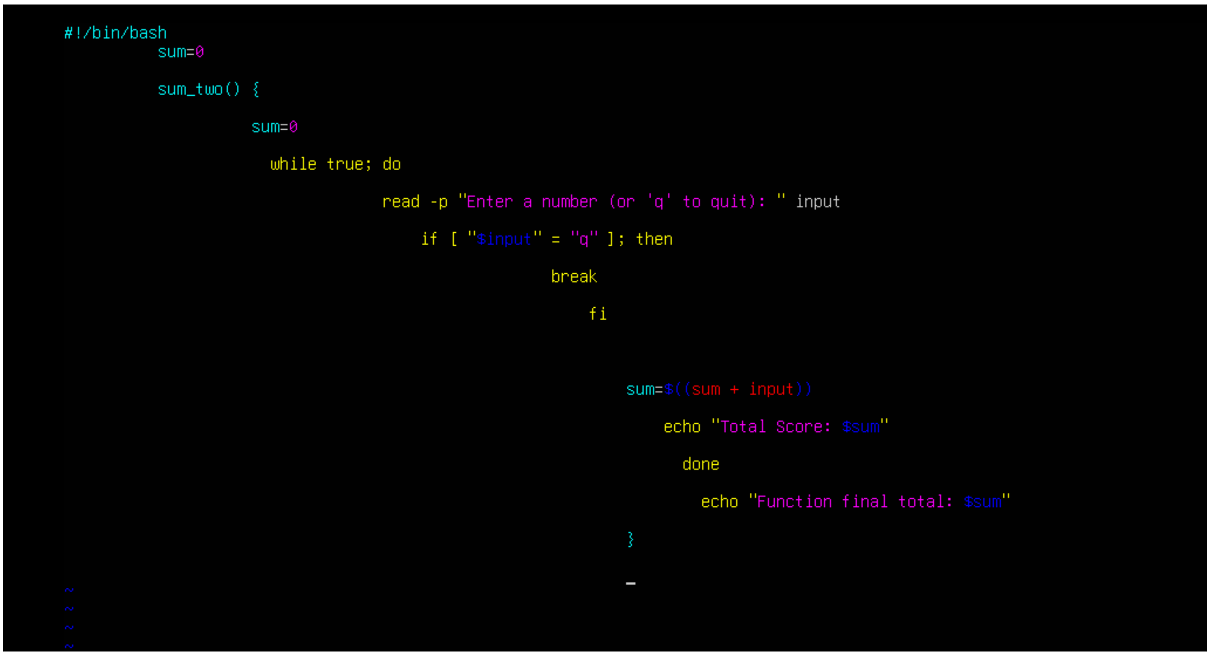
3. **Add the interactive summation function and demonstrate it**

- Re-open setup.sh in vim and append the function sum_two() which contains its own interactive while-loop:

```
# Function to accumulate scores interactively
sum_two() {
  sum=0
  while true; do
    read -p "Enter a number (or 'q' to quit): " input
    if [ "$input" = "q" ]; then
      break
    fi

    sum=$((sum + input))
    echo "Total Score: $sum"
  done
```

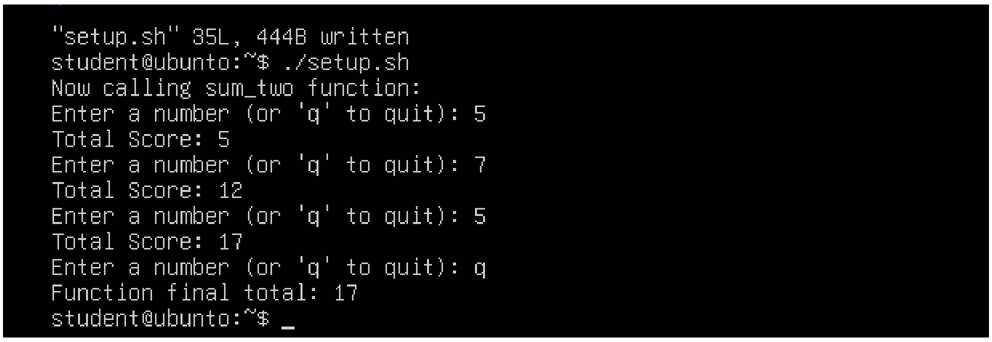
```
 echo "Function final total: $sum"
}

# Demonstrate the function
echo "Now calling sum_two function:"
sum_two
```

- Save and quit (:wq)



```
#!/bin/bash
sum=0

sum_two() {

sum=0

while true; do

read -p "Enter a number (or 'q' to quit): " input

if [ "$input" = "q" ]; then

break

fi


sum=$((sum + input))

echo "Total Score: $sum"

done

echo "Function final total: $sum"

}

_
~
~
~
~
```

./setup.sh



```
"setup.sh" 35L, 444B written
student@ubunto:~$ ./setup.sh
Now calling sum_two function:
Enter a number (or 'q' to quit): 5
Total Score: 5
Enter a number (or 'q' to quit): 7
Total Score: 12
Enter a number (or 'q' to quit): 5
Total Score: 17
Enter a number (or 'q' to quit): q
Function final total: 17
student@ubunto:~$ _
```

4. Add a function that takes two numeric arguments, sums them, and returns the result (echo)

```
# Function that sums two arguments and returns the result
sum_args() {
 a=$1
 b=$2
 return $((a + b))
}

# Demonstrate sum_args function
```

echo "Now demonstrating sum_args function:"
sum_args 3 4
result=$?
echo "sum_args(3,4) returned: $result"

- Save and quit (:wq)
- Run the script and capture the demonstration output:

```
#!/bin/bash
sum_args() {
  a=$1
  b=$2
  return $((a + b))
}

 Demonstrate sum_args function
 echo "Now demonstrating sum_args function:"
 sum_args 3 4
 result=$?
 echo "sum_args(3,4) returned: $result"
~
~
~
```

chmod +x setup.sh

./setup.sh

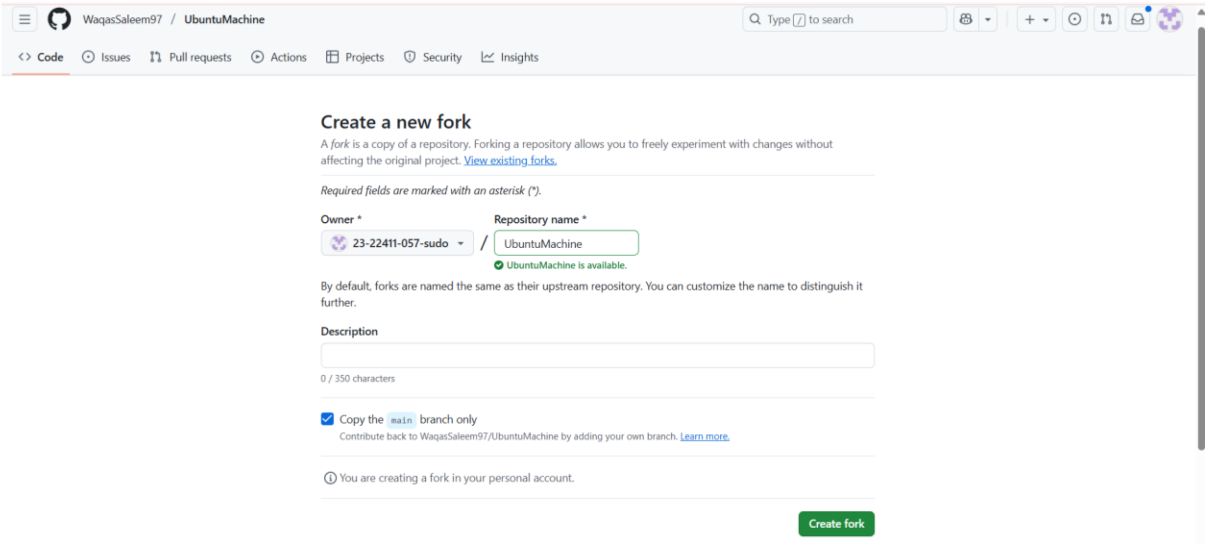
```
~
~
"setup.sh" 23L, 222B written
student@ubunto:~$ chmod +x setup.sh
student@ubunto:~$ ./setup.sh
./setup.sh: line 15: Demonstrate: command not found
Now demonstrating sum_args function:
sum_args(3,4) returned: 7
student@ubunto:~$
```

Task 14 – Codespaces GUI — fork repo, run start-desktop.sh, open VNC, stop GUI

Steps:

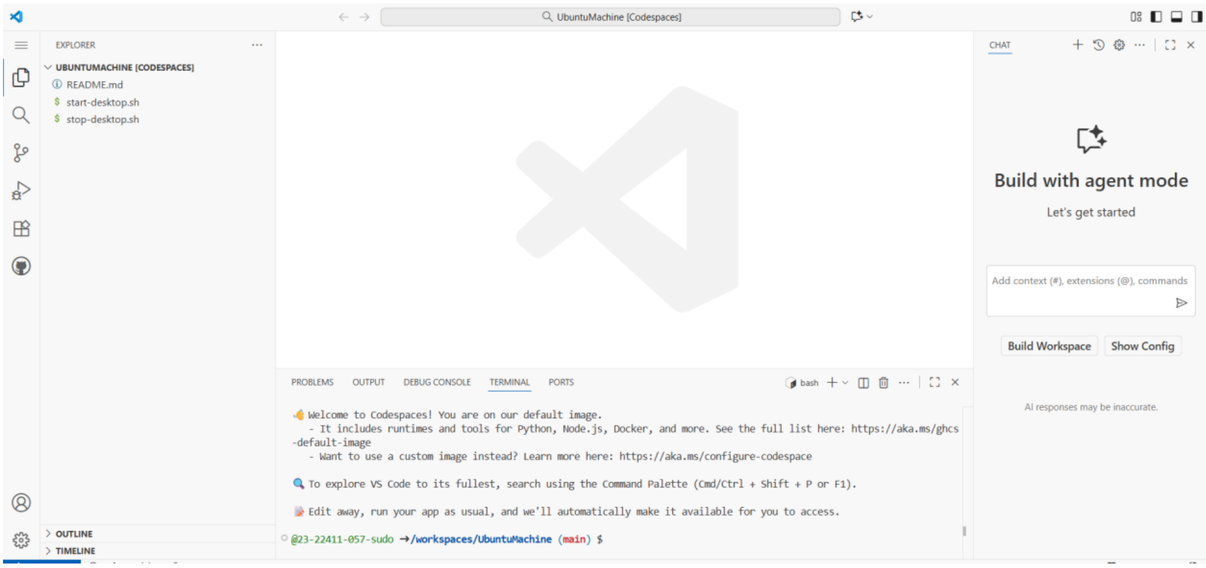
1. Fork the repository to your GitHub account

- Open the repo URL in your browser:
  - Ubuntu Machine
- Click "Fork" (top-right) and fork it to your account.



2. Open a Codespace on your fork

- In your forked repository on GitHub, click the green "Code" button → "Open with Codespaces" → "Create codespace on main" (or appropriate branch).
- Wait for the Codespace to initialize.



3. Verify the start script is present and executable (capture evidence)

- In the Codespace terminal list files in the repo root and show the start script and stop script exist:

ls -l start-desktop.sh stop-desktop.sh

- If not executable, make it executable:

chmod +x start-desktop.sh stop-desktop.sh

```
@23-22411-057-sudo →/workspaces/UbuntuMachine (main) $ ls -l start-desktop.sh stop-desktop.sh
-rwxrwxrwx 1 codespace root 1333 Nov  6 09:50 start-desktop.sh
-rwxrwxrwx 1 codespace root  428 Nov  6 09:50 stop-desktop.sh
@23-22411-057-sudo →/workspaces/UbuntuMachine (main) $ chmod +x start-desktop.sh stop-desktop.sh
@23-22411-057-sudo →/workspaces/UbuntuMachine (main) $ ls -l start-desktop.sh stop-desktop.sh
-rwxrwxrwx 1 codespace root 1333 Nov  6 09:50 start-desktop.sh
-rwxrwxrwx 1 codespace root  428 Nov  6 09:50 stop-desktop.sh
@23-22411-057-sudo →/workspaces/UbuntuMachine (main) $
```

4. Run the start script inside the Codespace terminal

- In the Codespace terminal run:

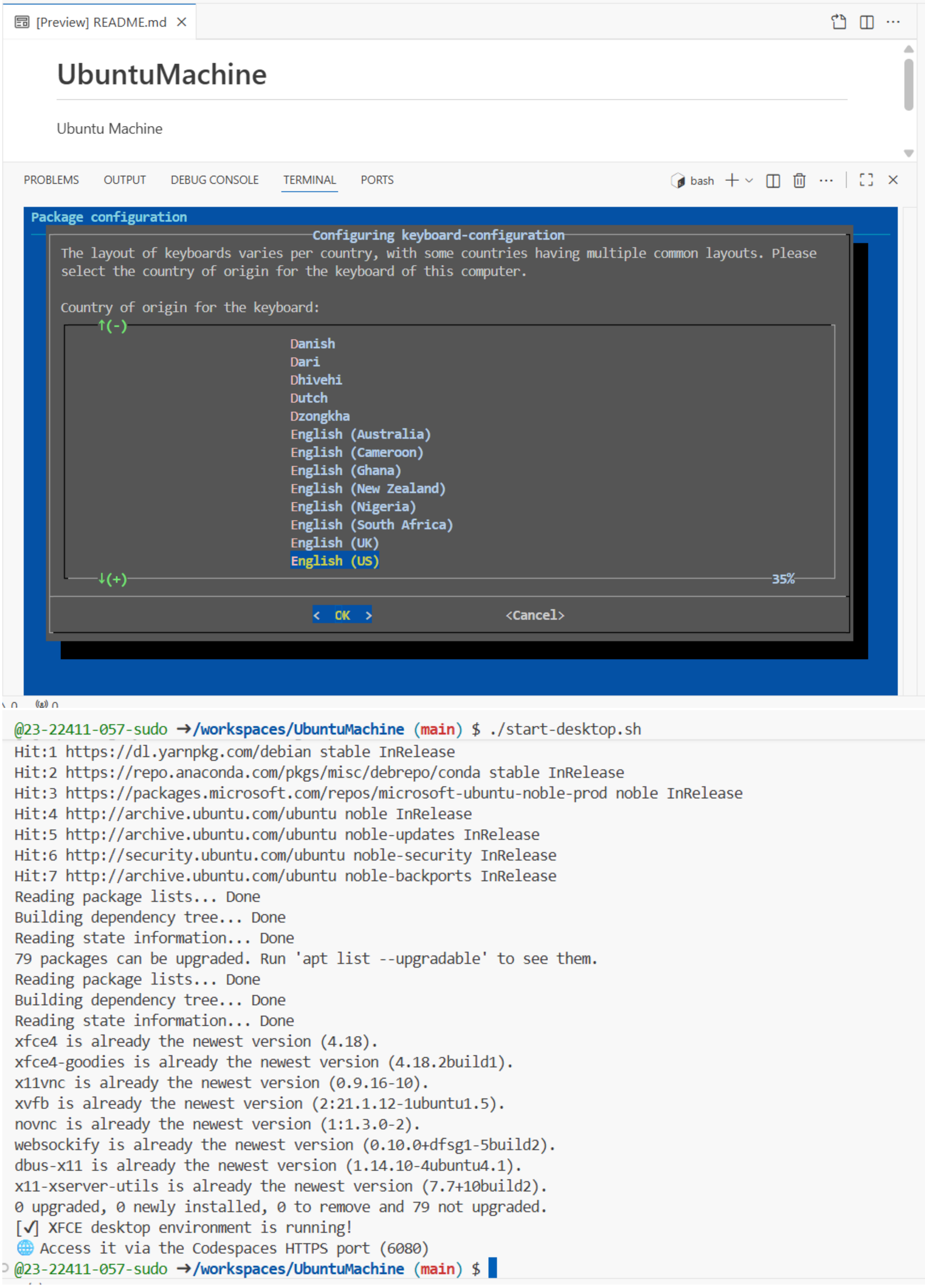
Ensure the start script is executable

chmod +x start-desktop.sh

Start the desktop GUI

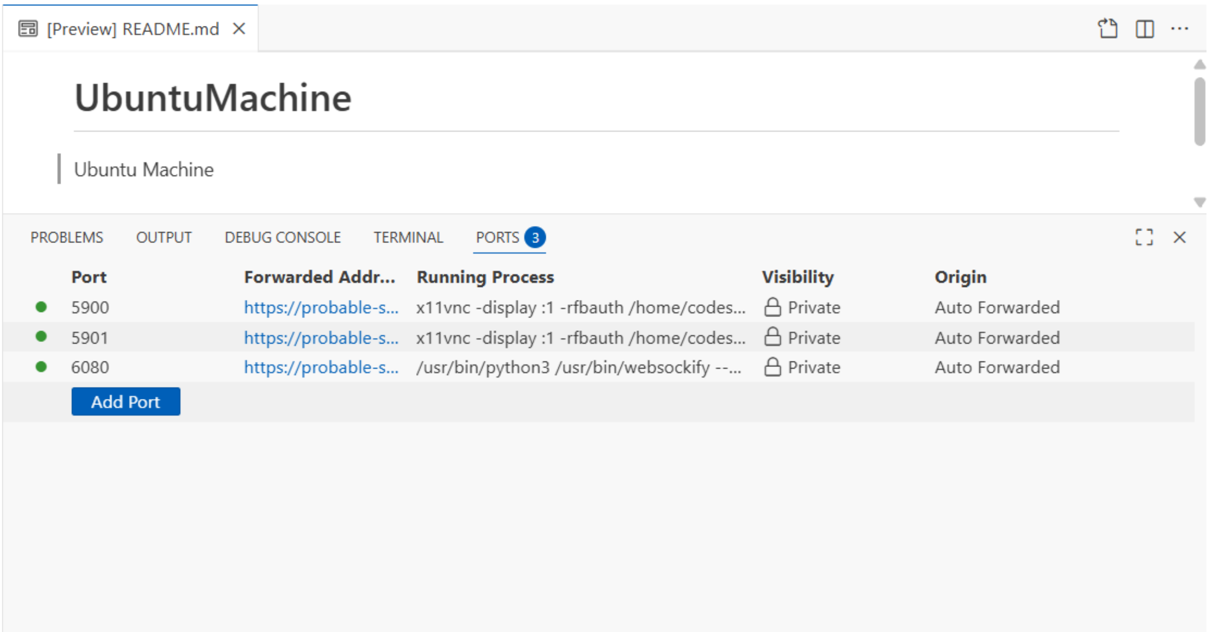
./start-desktop.sh

```
@23-22411-057-sudo →/workspaces/UbuntuMachine (main) $ chmod +x start-desktop.sh stop-desktop.sh
@23-22411-057-sudo →/workspaces/UbuntuMachine (main) $ ls -l start-desktop.sh stop-desktop.sh
-rwxrwxrwx 1 codespace root 1333 Nov  6 09:50 start-desktop.sh
-rwxrwxrwx 1 codespace root  428 Nov  6 09:50 stop-desktop.sh
@23-22411-057-sudo →/workspaces/UbuntuMachine (main) $ chmod +x start-desktop.sh
@23-22411-057-sudo →/workspaces/UbuntuMachine (main) $ ./start-desktop.sh
[*] Updating system and installing dependencies...
Get:1 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble InRelease [3600 B]
Get:2 https://repo.anaconda.com/pkgs/misc/debrepo/conda stable InRelease [3961 B]
Get:3 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble/main amd64 Packages [66.5 kB]
Get:4 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble/main all Packages [643 B]
Get:5 https://dl.yarnpkg.com/debian stable InRelease
Get:6 https://repo.anaconda.com/pkgs/misc/debrepo/conda stable/main amd64 Packages [4557 B]
Get:7 https://dl.yarnpkg.com/debian stable/main all Packages [11.8 kB]
Get:8 https://dl.yarnpkg.com/debian stable/main amd64 Packages [11.8 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [33.1 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1639 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
```



5. Verify forwarded ports in Codespaces (Ports view)

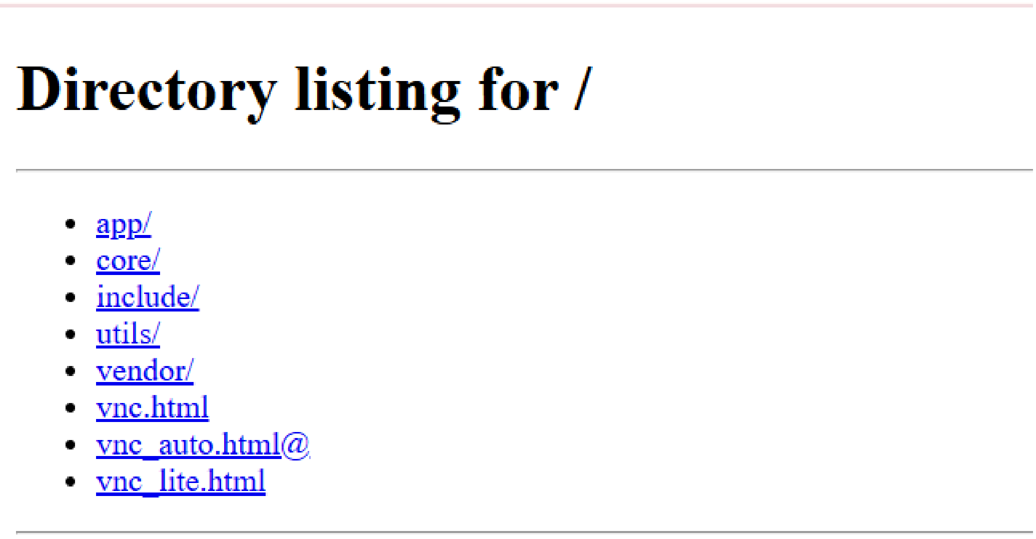
- Open the Codespaces "Ports" panel / view and confirm port 6080 is forwarded and visible.

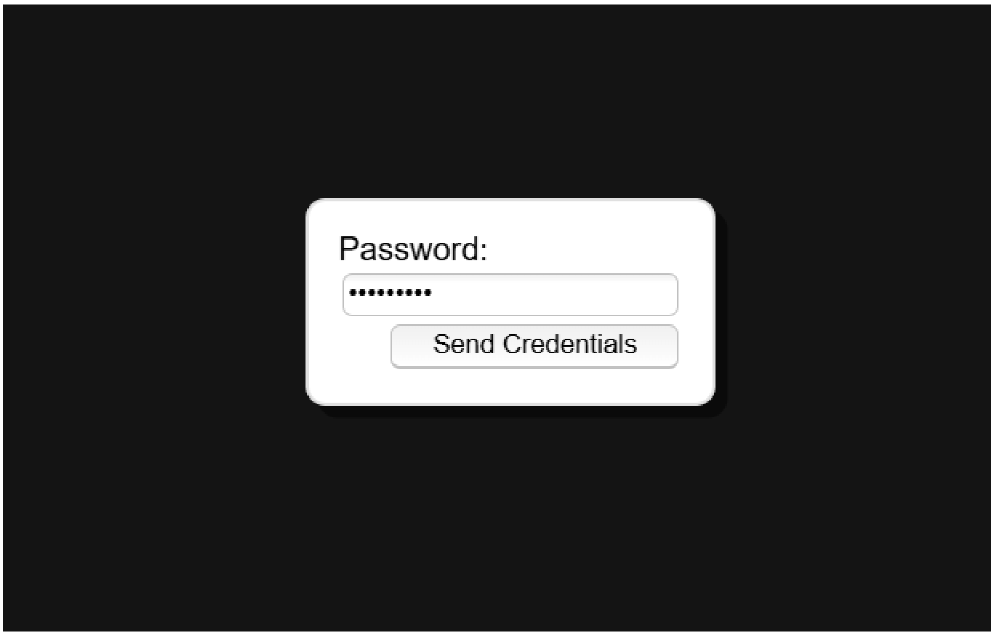


6. Open forwarded port 6080 and connect to VNC HTML page

- In the Codespaces UI, open the forwarded port's preview URL or copy the forwarded URL and open it in your browser.
- Visit the port 6080 address and click the vnc.html link.
- When prompted for a password enter:

Codespace





7. Stop the GUI

- When finished, return to the Codespace terminal and run:

./stop-desktop.sh

Exam Evaluation Questions

1. Group Management and Membership

Scenario:
Create groups and manage a user's primary and supplementary group memberships.

Steps:

1. Create groups g1, g2, and g3.

```
saira@ubunto:~$ sudo groupadd g1
[sudo] password for saira:
groupadd: group 'g1' already exists
saira@ubunto:~$ sudo groupadd g2
groupadd: group 'g2' already exists
saira@ubunto:~$ sudo groupadd g3
groupadd: group 'g3' already exists
```

2. Change examuser's primary group to g3 and add g1 and g2 as supplementary groups.

```
saira@ubunto:~$ sudo adduser examuser

info: Adding user `examuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `examuser' (1009) ...
info: Adding new user `examuser' (1009) with group `examuser (1009)' ...
info: Creating home directory `/home/examuser' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for examuser
Enter the new value, or press ENTER for the default
        Full Name []: examuser
        Room Number []: 12
        Work Phone []: 2222
        Home Phone []: 33333
        Other []: n/a
Is the information correct? [Y/n] y
info: Adding new user `examuser' to supplemental / extra groups `users' ...
info: Adding user `examuser' to group `users' ...
saira@ubunto:~$ sudo usermod -g g3 examuser


saira@ubunto:~$ sudo usermod -aG g1,g2 examuser

saira@ubunto:~$ _
```

3. Show the final id and /etc/group lines that prove the changes.

```
saira@ubunto:~$ id examuser

uid=1009(examuser) gid=1008(g3) groups=1008(g3),100(users),1006(g1),1007(g2)
saira@ubunto:~$
saira@ubunto:~$ grep examuser /etc/group
users:x:100:student,examuser
g1:x:1006:examuser
g2:x:1007:examuser
examuser:x:1009:
saira@ubunto:~$
saira@ubunto:~$
```

2. Ownership and Permission Tasks

Scenario:
Demonstrate ownership changes and apply both symbolic and numeric permission changes.

Steps:

1. Create workspace/secret.txt, change its owner to examuser and group to g1.

```
saira@ubunto:~$
saira@ubunto:~$ mkdir -p workspace
saira@ubunto:~$ touch workspace/secret.txt
saira@ubunto:~$ sudo chown examuser:g1 workspace/secret.txt


saira@ubunto:~$ ls -l workspace/secret.txt

-rw-rw-r-- 1 examuser g1 0 Nov  6 11:15 workspace/secret.txt
saira@ubunto:~$
```

2. Remove all permissions for group and others using a symbolic command, then using a numeric command to achieve the same result.

```
examuser@ubuntu:~$ sudo chown examuser:g1 ~/workspace/secret.txt
[sudo] password for examuser:
examuser is not in the sudoers file.
examuser@ubuntu:~$ chmod go-rwx workspace/secret.txt
examuser@ubuntu:~$ chmod 700 workspace/secret.txt
examuser@ubuntu:~$ ls -l workspace/secret.txt
-rwx------ 1 examuser g3 0 Nov  6 11:20 workspace/secret.txt
examuser@ubuntu:~$
```

3. Show ls -l for the file after each change to document the permission bits.

```
examuser@ubuntu:~$ ls -l workspace/secret.txt
-rwx------ 1 examuser g3 0 Nov  6 11:20 workspace/secret.txt
examuser@ubuntu:~$
examuser@ubuntu:~$
```

3. Pipes, Grep, and Redirection Practice

Scenario:
Filter system logs and save results using redirection and piping.

Steps:

1. Use grep (or journalctl where applicable) with a pipe to find lines containing "error" or "fail" and show the first 20 results.

```
saira@ubuntu:~$ sudo grep -i -E 'error|fail' /var/log/syslog | head -n 20

2025-11-03T14:13:17.892120+00:00 ubuntu multipathd[500]: sda: failed to get udev uid: No data available
2025-11-03T14:13:17.892330+00:00 ubuntu multipath: sda: failed to get sysfs uid: No such file or directory
2025-11-03T14:13:17.892340+00:00 ubuntu multipath: sda: failed to get sgio uid: No such file or directory
2025-11-03T14:13:17.892391+00:00 ubuntu multipathd[500]: sda: failed to get udev uid: No data available
2025-11-03T14:13:17.905911+00:00 ubuntu multipathd[500]: sda: failed to get path uid
2025-11-03T14:13:17.905956+00:00 ubuntu multipathd[500]: uevent trigger error
2025-11-03T14:13:17.930779+00:00 ubuntu multipath: sda: failed to get sysfs uid: No such file or directory
2025-11-03T14:13:17.930841+00:00 ubuntu multipath: sda: failed to get sgio uid: No such file or directory
2025-11-03T14:13:17.930854+00:00 ubuntu multipathd[500]: sda: failed to get udev uid: No data available
2025-11-03T14:13:17.930864+00:00 ubuntu multipathd[500]: sda: failed to get path uid
2025-11-03T14:13:17.930879+00:00 ubuntu multipathd[500]: uevent trigger error
2025-11-03T14:13:17.970533+00:00 ubuntu systemd[1]: apport-autoreport.path - Process error reports when automatic reporting is enabled (file watch) was skipped
because of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
2025-11-03T14:13:17.970545+00:00 ubuntu systemd[1]: apport-autoreport.timer - Process error reports when automatic reporting is enabled (timer based) was skippe
d because of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
2025-11-03T14:13:17.970625+00:00 ubuntu systemd[1]: Started update-notifier-download.timer - Download data for packages that failed at package install time.
2025-11-03T14:13:17.971075+00:00 ubuntu systemd[1]: Starting grub-initrd-fallback.service - GRUB failed boot detection...
2025-11-03T14:13:17.971815+00:00 ubuntu kernel: unchecked MSR access error: RDMSR from 0x852 at rIP: 0xffffffffb50c69f7 (native_read_msr+0x7/0x50)
2025-11-03T14:13:17.972031+00:00 ubuntu kernel: ACPI: _OSC evaluation for CPUs failed, trying _PDC
2025-11-03T14:13:17.989825+00:00 ubuntu kernel: pci 0000:00:15.3: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989826+00:00 ubuntu kernel: pci 0000:00:15.4: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989826+00:00 ubuntu kernel: pci 0000:00:15.5: bridge window [io  size 0x1000]: failed to assign
saira@ubuntu:~$
saira@ubuntu:~$
```

2. Save the filtered results to a file ~/logs/errors.txt using overwrite, then append additional matching lines using append redirection.

```
saira@ubuntu:~$ mkdir -p ~/logs
saira@ubuntu:~$ sudo grep -i -E 'error|fail' /var/log/syslog | head -n 20 > ~/logs/errors.txt
saira@ubuntu:~$ sudo grep -i -E 'error|fail' /var/log/syslog | tail -n +21 >> ~/logs/errors.txt

grep: /var/log/syslog: binary file matches
saira@ubuntu:~$
saira@ubuntu:~$
```

3. Use a pager to view the saved file.

```
2025-11-03T14:13:17.892330+00:00 ubuntu multipath: sda: failed to get sysfs uid: No such file or directory
2025-11-03T14:13:17.892340+00:00 ubuntu multipath: sda: failed to get sgio uid: No such file or directory
2025-11-03T14:13:17.892391+00:00 ubuntu multipathd[500]: sda: failed to get udev uid: No data available
2025-11-03T14:13:17.905911+00:00 ubuntu multipathd[500]: sda: failed to get path uid
2025-11-03T14:13:17.905956+00:00 ubuntu multipathd[500]: uevent trigger error
2025-11-03T14:13:17.930779+00:00 ubuntu multipath: sda: failed to get sysfs uid: No such file or directory
2025-11-03T14:13:17.930841+00:00 ubuntu multipath: sda: failed to get sgio uid: No such file or directory
2025-11-03T14:13:17.930854+00:00 ubuntu multipathd[500]: sda: failed to get udev uid: No data available
2025-11-03T14:13:17.930864+00:00 ubuntu multipathd[500]: sda: failed to get path uid
2025-11-03T14:13:17.930879+00:00 ubuntu multipathd[500]: uevent trigger error
2025-11-03T14:13:17.970533+00:00 ubuntu systemd[1]: apport-autoreport.path - Process error reports when automatic reporting is enabled (file watch) was skipped
because of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
2025-11-03T14:13:17.970545+00:00 ubuntu systemd[1]: apport-autoreport.timer - Process error reports when automatic reporting is enabled (timer based) was skippe
d because of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
2025-11-03T14:13:17.970625+00:00 ubuntu systemd[1]: Started update-notifier-download.timer - Download data for packages that failed at package install time.
2025-11-03T14:13:17.971075+00:00 ubuntu systemd[1]: Starting grub-initrd-fallback.service - GRUB failed boot detection...
2025-11-03T14:13:17.971815+00:00 ubuntu kernel: unchecked MSR access error: RDMSR from 0x852 at rIP: 0xffffffffb50c69f7 (native_read_msr+0x7/0x50)
2025-11-03T14:13:17.972031+00:00 ubuntu kernel: ACPI: _OSC evaluation for CPUs failed, trying _PDC
2025-11-03T14:13:17.989825+00:00 ubuntu kernel: pci 0000:00:15.3: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989826+00:00 ubuntu kernel: pci 0000:00:15.4: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989826+00:00 ubuntu kernel: pci 0000:00:15.5: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989827+00:00 ubuntu kernel: pci 0000:00:15.6: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989828+00:00 ubuntu kernel: pci 0000:00:15.7: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989828+00:00 ubuntu kernel: pci 0000:00:16.3: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989829+00:00 ubuntu kernel: pci 0000:00:16.4: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989830+00:00 ubuntu kernel: pci 0000:00:16.5: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989830+00:00 ubuntu kernel: pci 0000:00:16.6: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989831+00:00 ubuntu kernel: pci 0000:00:16.7: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989832+00:00 ubuntu kernel: pci 0000:00:17.3: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989832+00:00 ubuntu kernel: pci 0000:00:17.4: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989833+00:00 ubuntu kernel: pci 0000:00:17.5: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989834+00:00 ubuntu kernel: pci 0000:00:17.6: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989834+00:00 ubuntu kernel: pci 0000:00:17.7: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989835+00:00 ubuntu kernel: pci 0000:00:18.2: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989836+00:00 ubuntu kernel: pci 0000:00:18.3: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989836+00:00 ubuntu kernel: pci 0000:00:18.4: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989837+00:00 ubuntu kernel: pci 0000:00:18.5: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989838+00:00 ubuntu kernel: pci 0000:00:18.6: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989838+00:00 ubuntu kernel: pci 0000:00:18.7: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989839+00:00 ubuntu kernel: pci 0000:00:18.7: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989840+00:00 ubuntu kernel: pci 0000:00:18.6: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989840+00:00 ubuntu kernel: pci 0000:00:18.5: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989841+00:00 ubuntu kernel: pci 0000:00:18.4: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989841+00:00 ubuntu kernel: pci 0000:00:18.3: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989842+00:00 ubuntu kernel: pci 0000:00:18.2: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989843+00:00 ubuntu kernel: pci 0000:00:17.7: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989844+00:00 ubuntu kernel: pci 0000:00:17.6: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989844+00:00 ubuntu kernel: pci 0000:00:17.5: bridge window [io  size 0x1000]: failed to assign
2025-11-03T14:13:17.989845+00:00 ubuntu kernel: pci 0000:00:17.4: bridge window [io  size 0x1000]: failed to assign
saira@ubuntu:~$
```

4. Script: Variables, Command Substitution, File & Dir Checks

Scenario:
Build and run a script incrementally that demonstrates variables, command substitution, and filesystem checks.

Steps:

1. Create setup.sh with a shebang and a variable var1 that you echo.

```
#!/bin/bash

var1="Hello, this is var1"

echo $var1
~
~
~
```

```
"setup.sh" [New] 6L, 78B written
saira@ubuntu:~$ chmod +x setup.sh
saira@ubuntu:~$ ./setup.sh
Hello, this is var1
saira@ubuntu:~$
```

2. Append command substitution that stores ls -l output into a variable and echo it.

```
#!/bin/bash

var1="Hello, this is var1"

echo $var1
 allfiles=$(ls -l)

 echo "$allfiles"_
```

```
saira@ubunto:~$ ./setup.sh
Hello, this is var1
total 384
-rw-rw-r-- 1 saira saira    476 Nov  4 10:34 apt_update_vs_upgrade.md
drwxr-xr-x 2 saira saira   4096 Nov  4 11:15 Desktop
drwxr-xr-x 2 saira saira   4096 Nov  4 11:15 Documents
drwxr-xr-x 2 saira saira   4096 Nov  4 11:15 Downloads
-rw-rw-r-- 1 saira saira      0 Nov  6 08:28 journal_errors.txt
drwxrwxr-x 2 saira saira   4096 Nov  4 12:54 Lab5
drwxrwxr-x 2 saira saira   4096 Nov  6 11:30 logs
drwxr-xr-x 2 saira saira   4096 Nov  4 11:15 Music
drwxr-xr-x 2 saira saira   4096 Nov  4 11:15 Pictures
drwxr-xr-x 2 saira saira   4096 Nov  4 11:15 Public
-rwxrwxr-x 1 saira saira    126 Nov  6 11:36 setup.sh
drwx------ 3 saira saira   4096 Nov  4 10:46 snap
-rw-rw-r-- 1 saira saira 328889 Nov  6 08:27 syslog_systemd.txt
drwxr-xr-x 2 saira saira   4096 Nov  4 11:15 Templates
drwxrwxr-t 2 saira saira   4096 Nov  4 11:15 thinclient_drives
drwxr-xr-x 2 saira saira   4096 Nov  4 11:15 Videos
drwxrwxr-x 2 saira saira   4096 Nov  6 11:15 workspace
saira@ubunto:~$
```

3. Append directory and file checks that create dir1 and dir1/file2 if missing, and display their final permissions.

```
#!/bin/bash

var1="Hello, this is var1"

echo $var1
     if [ ! -d "dir1" ]; then

                 mkdir dir1

     fi


    # Check if dir1/file2 exists, create if missing

      if [ ! -f "dir1/file2" ]; then

       touch dir1/file2

          fi


          # Show final permissions

           ls -l dir1
~
~
```

```
~
"setup.sh" 27L, 338B written
saira@ubunto:~$ ./setup.sh
Hello, this is var1
total 0
-rw-rw-r-- 1 saira saira 0 Nov  6 11:40 file2
saira@ubunto:~$
```

5. Script: Comparisons and String Tests

Scenario:
Incrementally add numeric and string comparison tests to a script and show both true/false cases.

Steps:

1. Overwrite setup.sh to set num=$1 and str=$2, and add an -eq test showing true and false examples.

```
#!/bin/bash
    num=$1
    str=$2
      if [ "$num" -eq 10 ]; then
                echo "$num is equal to 10"
         else
                   echo "$num is not equal to 10"
      fi_
~
~
```

```
"setup.sh" 15L, 187B written
saira@ubunto:~$ chmod +x setup.sh
saira@ubunto:~$ ./setup.sh 10 hello
10 is equal to 10
saira@ubunto:~$ ./setup.sh 5 hello
5 is not equal to 10
saira@ubunto:~$ _
```

2. Append -ne, -gt, -lt, -ge, and -le tests and demonstrate at least one true and one false invocation for each.

```
            fi
            if [ "$num" -ne 10 ]; then
                   echo "$num is not equal to 10"
              else
                     echo "$num is equal to 10"
            fi
           if [ "$num" -gt 7 ]; then
                  echo "$num is greater than 7"
             else
                    echo "$num is not greater than 7"
           fi
          if [ "$num" -lt 20 ]; then
      echo "$num is less than 20"
   else
      echo "$num is not less than 20"
          fi
       if [ "$num" -ge 10 ]; then
      echo "$num is greater than or equal to 10"
   else
      echo "$num is less than 10"
       fi
    if [ "$num" -le 15 ]; then
      echo "$num is less than or equal to 15"
   else
      echo "$num is greater than 15"
    fi_
   -- INSERT --
```

```
 fi
"setup.sh" 62L, 826B written
saira@ubunto:~$ ./setup.sh 10 test
10 is equal to 10
10 is equal to 10
10 is greater than 7
10 is less than 20
10 is greater than or equal to 10
10 is less than or equal to 15
saira@ubunto:~$ ./setup.sh 5 test
5 is not equal to 10
5 is not equal to 10
5 is not greater than 7
5 is less than 20
5 is less than 10
5 is less than or equal to 15
saira@ubunto:~$
```

3. Append string equality (=) and inequality (!=) checks and a -z (zero-length) test for the second argument, demonstrating true/false cases.

```
        fi
     if [ "$num" -ge 10 ]; then
    echo "$num is greater than or equal to 10"
else
    echo "$num is less than 10"
     fi
 if [ "$num" -le 15 ]; then
    echo "$num is less than or equal to 15"
else
    echo "$num is greater than 15"
fi
 if [ "$str" = "hello" ]; then
            echo "$str is equal to hello"
     else
                echo "$str is not equal to hello"
 fi
 if [ "$str" != "world" ]; then
            echo "$str is not equal to world"
     else
                echo "$str is equal to world"
 fi
 if [ -z "$str" ]; then
            echo "Second argument is empty"
     else
                echo "Second argument is not empty"
 fi
-- INSERT --                                                   91,4          Bot
```

```
"setup.sh" 91L, 1208B written
saira@ubunto:~$ ./setup.sh 10 test
10 is equal to 10
10 is equal to 10
10 is greater than 7
10 is less than 20
10 is greater than or equal to 10
10 is less than or equal to 15
test is not equal to hello
test is not equal to world
Second argument is not empty
saira@ubunto:~$ ./setup.sh 5 test
5 is not equal to 10
5 is not equal to 10
5 is not greater than 7
5 is less than 20
5 is less than 10
5 is less than or equal to 15
test is not equal to hello
test is not equal to world
Second argument is not empty
saira@ubunto:~$ _
```

6. Script: For Loop and Argument Handling

Scenario:
Write a script that prints all provided arguments and demonstrate correct handling of quoted multi-word arguments.

Steps:

1. Create/overwrite setup.sh to print every argument using "$@" in a for loop and save the file.

```
#!/bin/bash
   for arg in "$@"
   do
       echo "$arg"
  done_
~
```

2. Run the script with mixed single and quoted multi-word arguments and capture the output showing each argument on its own line.

```
"setup.sh" 10L, 83B written
saira@ubunto:~$ chmod +x setup.sh
saira@ubunto:~$ ./setup.sh apple "green banana" orange 'red apple'
apple
green banana
orange
red apple
saira@ubunto:~$ _
```

7. Script: While Loop Summation and Functions

Scenario:
Implement an interactive or non-interactive summation function and a demonstrated function that returns a numeric result.

Steps:

1. Write an interactive while-loop that accumulates numbers until q is entered and shows running totals.

```
#!/bin/bash
total=0
while true; do
read -p "Enter a number (or 'q' to quit): " num
if [ "$num" = "q" ]; then
break
fi
# Check if input is a number
if [[ "$num" =~ ^[0-9]+$ ]]; then
total=$((total + num))
echo "Running total: $total"
else
echo "Invalid input, enter a number or 'q' to quit."
fi
done
echo "Final total: $total"
```

```
"setup.sh" 32L, 680B written
saira@ubunto:~$ chmod +x setup.sh
saira@ubunto:~$ ./setup.sh
Enter a number (or 'q' to quit): 5
Running total: 5
Enter a number (or 'q' to quit): 6
Running total: 11
Enter a number (or 'q' to quit): 5
Running total: 16
Enter a number (or 'q' to quit): q
Final total: 16
saira@ubunto:~$ _
```

2. Add a function that accepts two numeric arguments, returns their sum, and demonstrate capturing its result in a variable.

```
#!/bin/bash
        sum_two_numbers() {
                       local a=$1
                           local b=$2
                               echo $((a + b))
                    }result=$(sum_two_numbers 5 7)
                  echo "Sum of 5 and 7 is: $result"
~
~
```

```
~
"setup.sh" 17L, 237B written
saira@ubunto:~$ ./setup.sh
Sum of 5 and 7 is: 12
saira@ubunto:~$ _
```